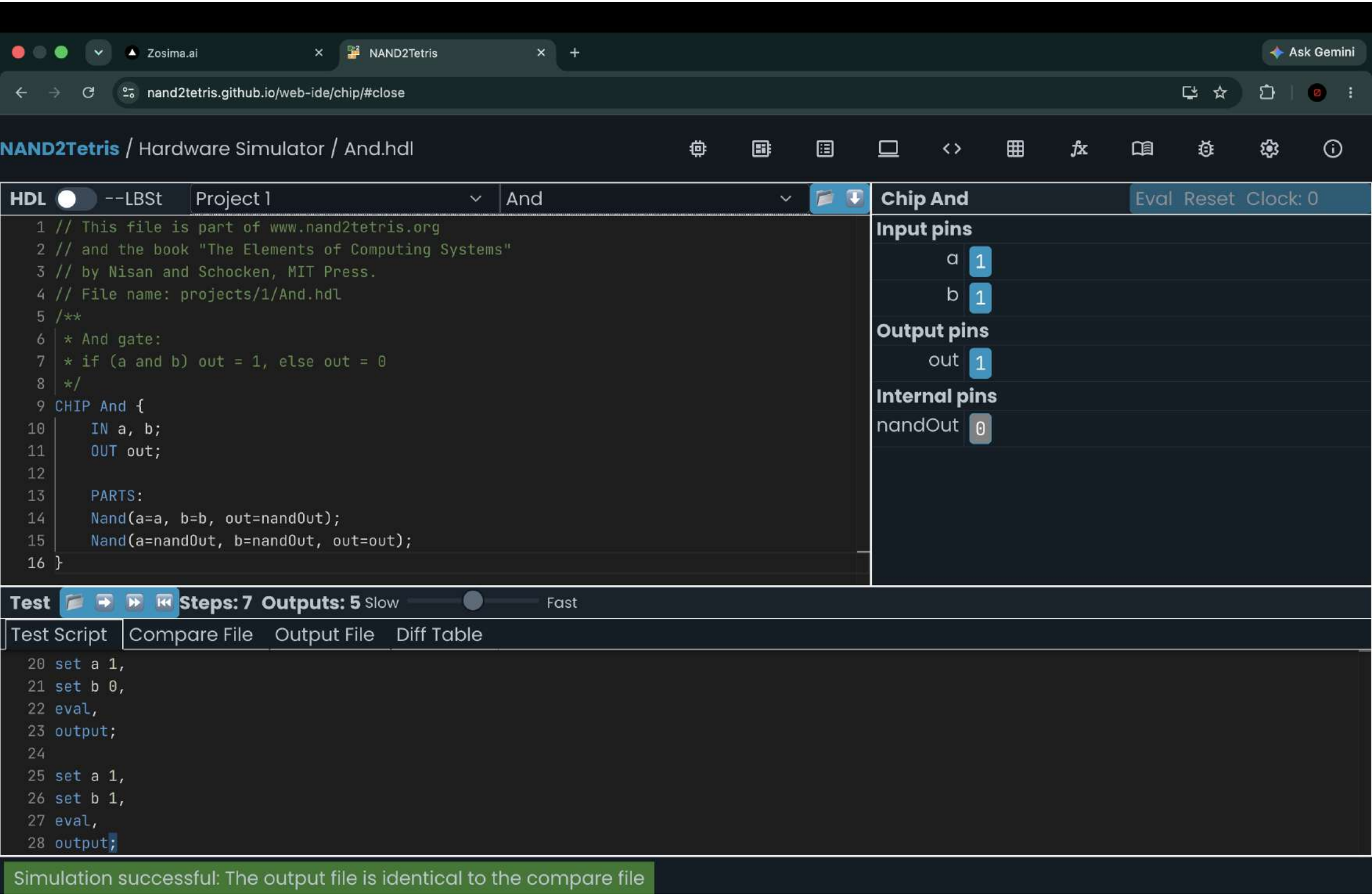
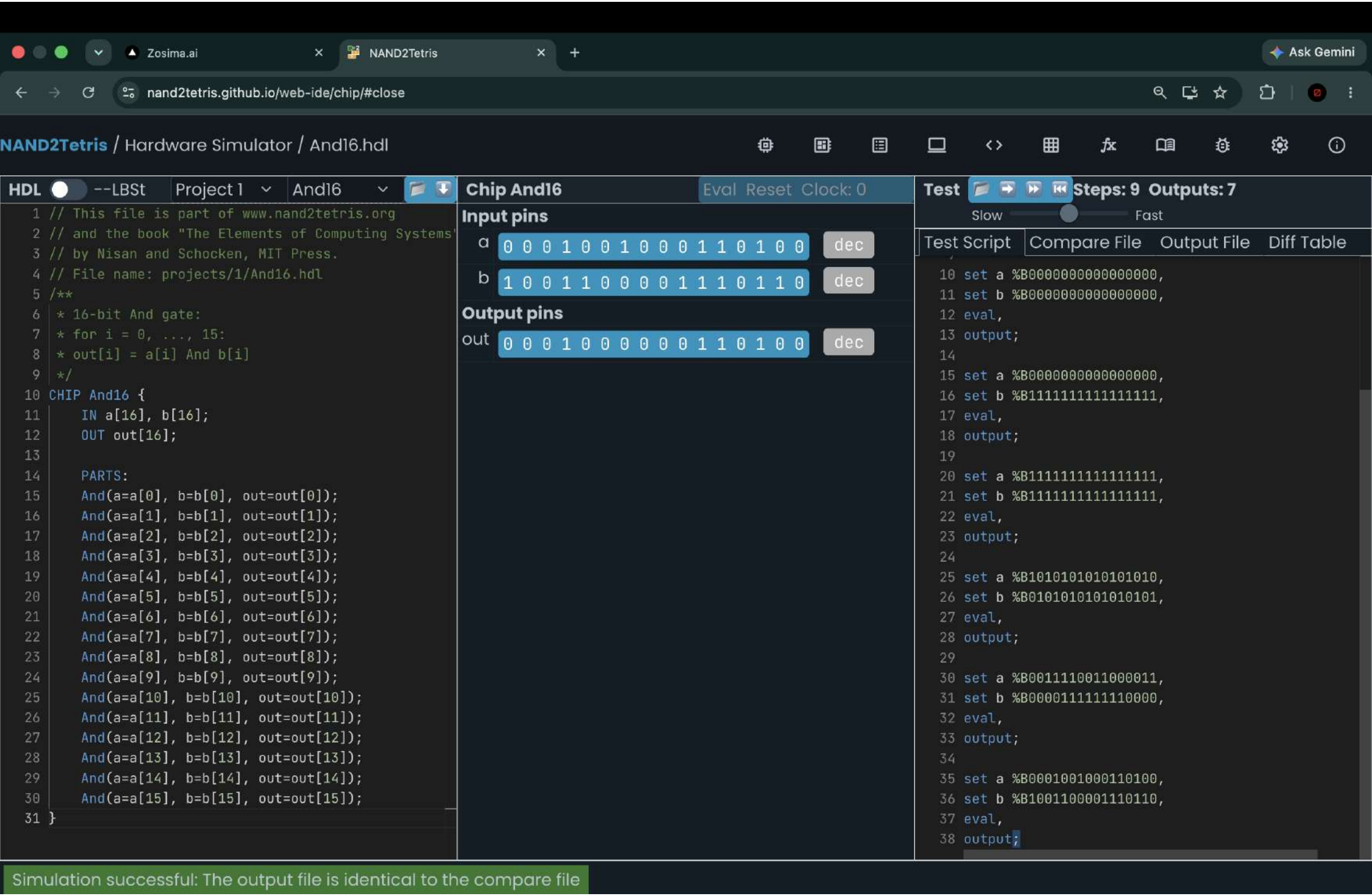
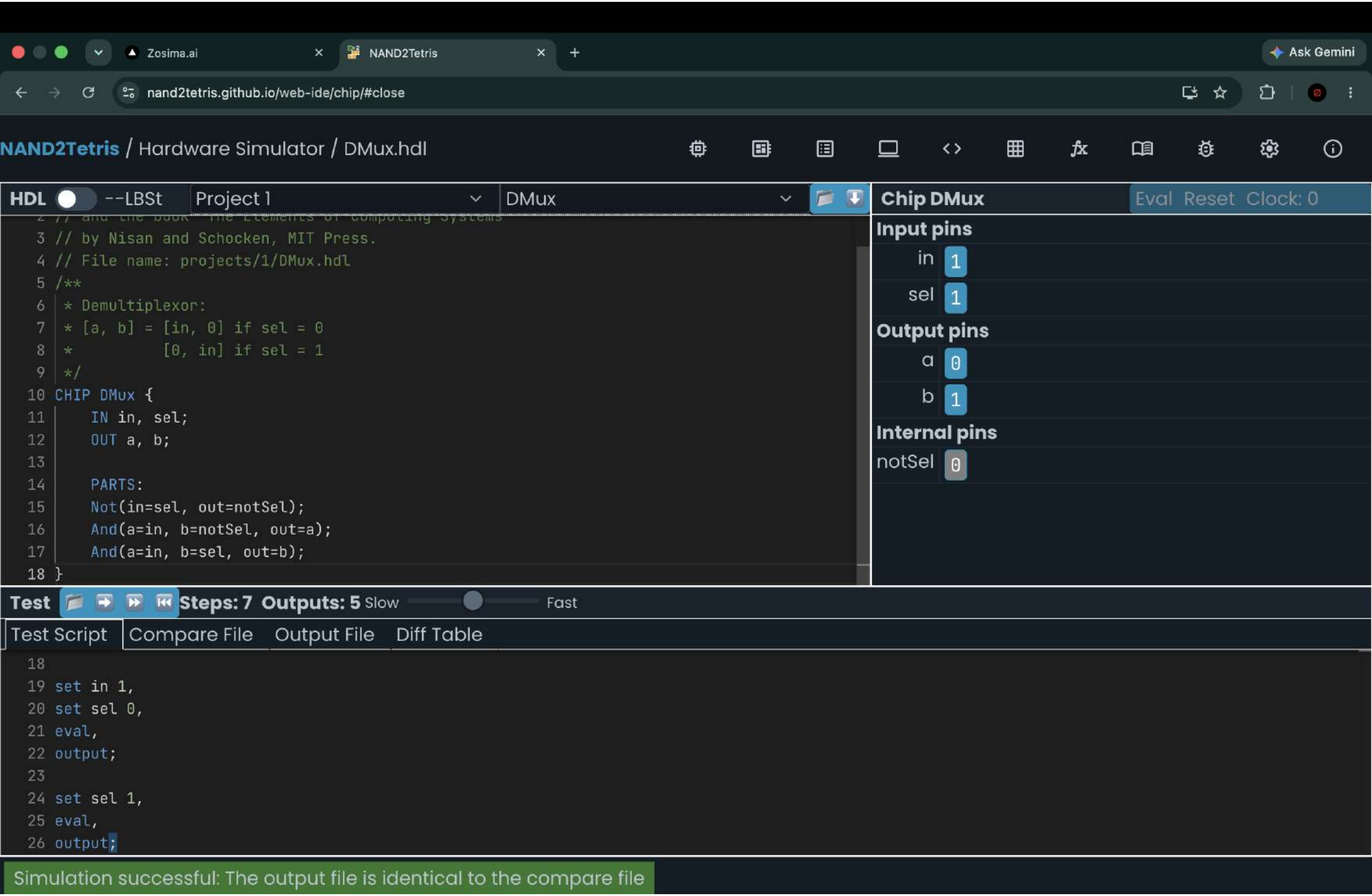
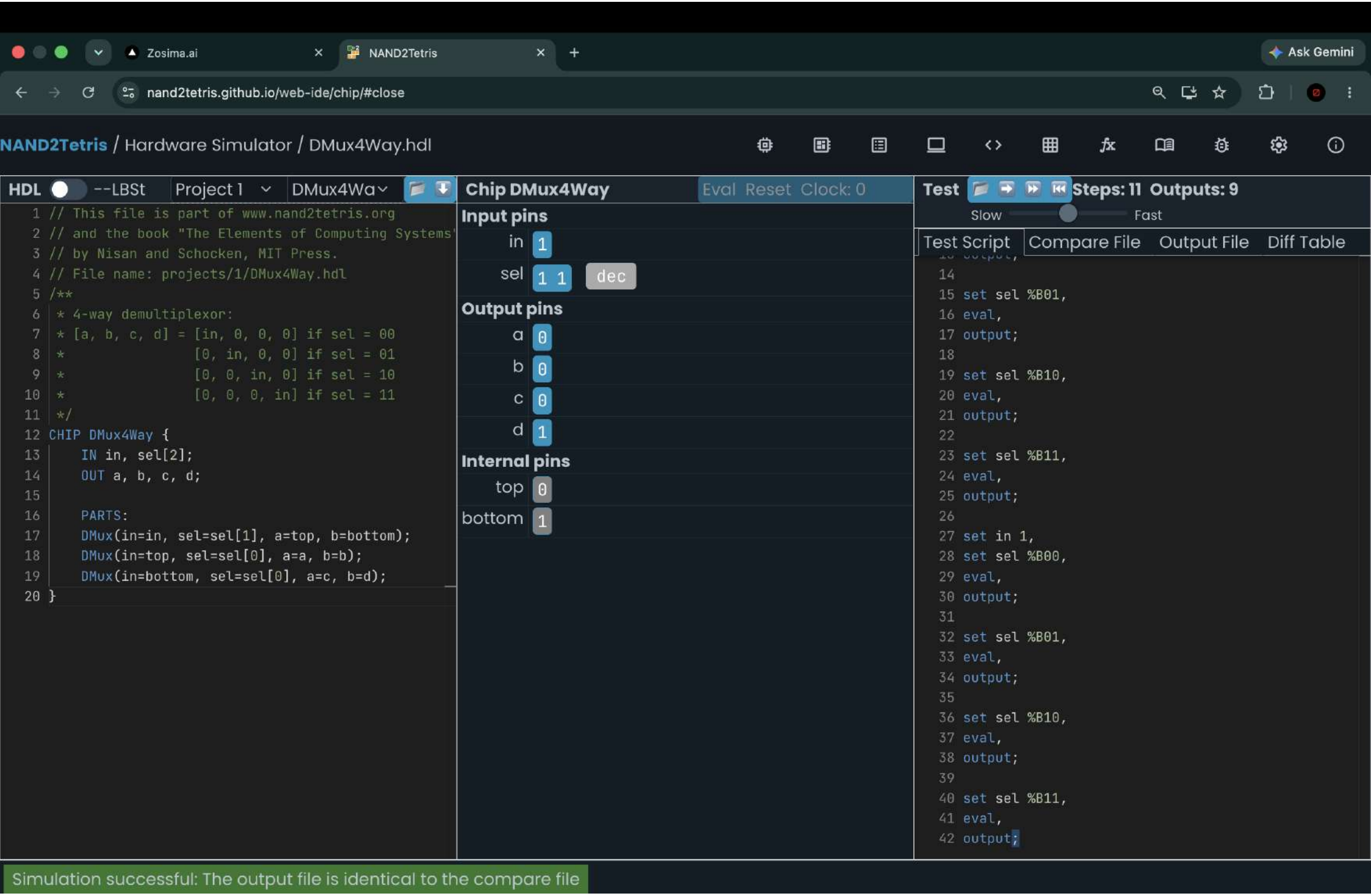
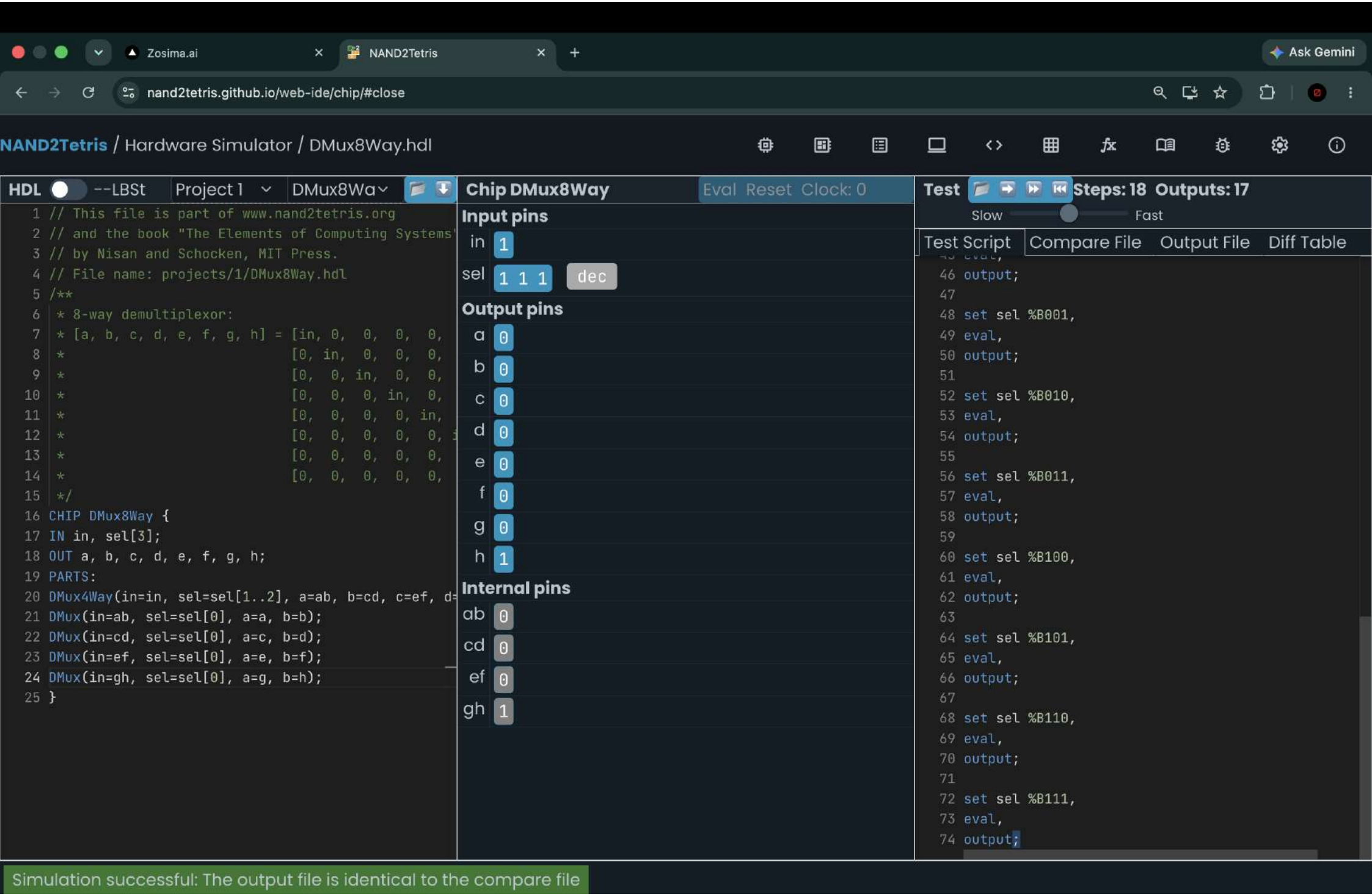
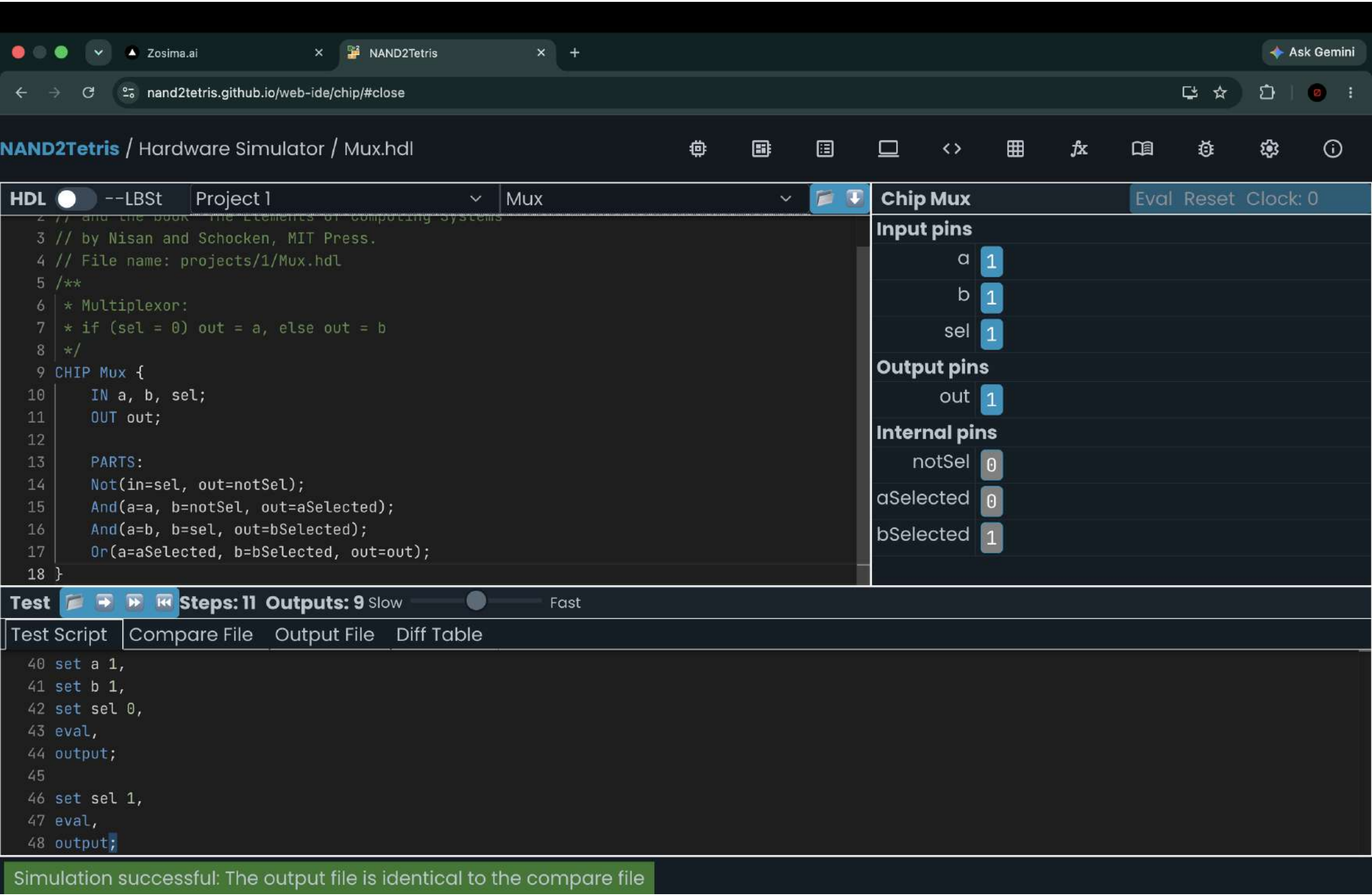

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NAND2Tetris / Hardware Simulator / And.hdl
HDL --LBSt
Project 1
And
1 // This file is part of www.nand2tetris.org
2 // and the book "The Elements of Computing Systems"
3 // by Nisan and Schocken, MIT Press.
4 // File name: projects/1/And.hdl
5 /**
6 * And gate:
7 * if (a and b) out = 1, else out = 0
8 */
9 CHIP And {
10 IN a, b;
11 OUT out;
12
13 PARTS:
14 Nand(a=a, b=b, out=nandOut);
15 Nand(a=nandOut, b=nandOut, out=out);
16 }
Chip And
Eval Reset Clock: 0
Input pins
a 1
b 1
Output pins
out 1
Internal pins
nandOut 0
Test
Steps: 7 Outputs: 5
Slow
Fast
Test Script
Compare File
Output File
Diff Table
20 set a 1,
21 set b 0,
22 eval,
23 output;
24
25 set a 1,
26 set b 1,
27 eval,
28 output;
Simulation successful: The output file is identical to the compare file

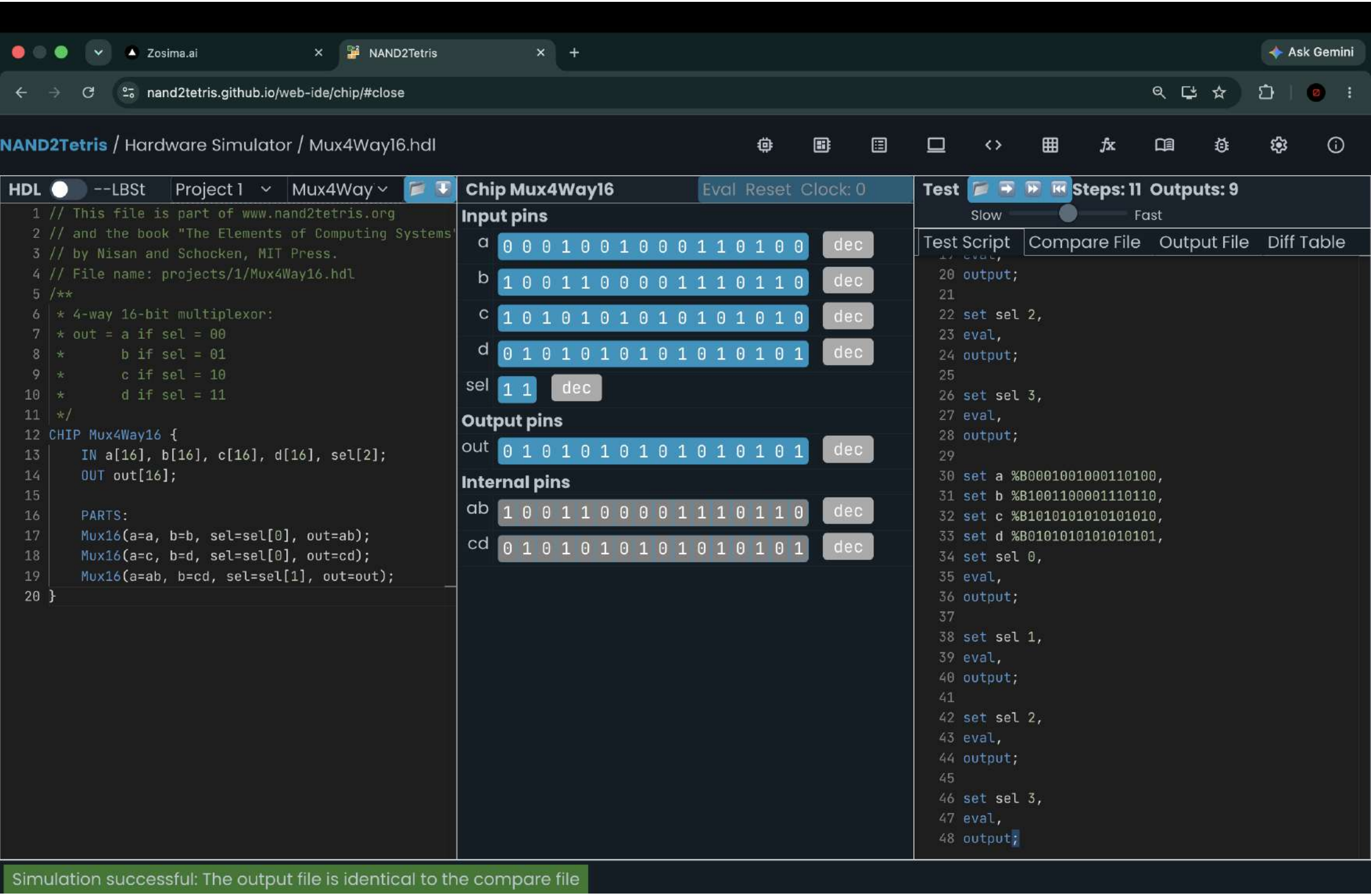

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NAND2Tetris / Hardware Simulator / And16.hdl
HDL --LBSt Project 1 And16
// This file is part of www.nand2tetris.org
// and the book "The Elements of Computing Systems"
// by Nisan and Schocken, MIT Press.
// File name: projects/1/And16.hdl
/**
* 16-bit And gate:
* for i = 0, ..., 15:
* out[i] = a[i] And b[i]
*/
CHIP And16 {
IN a[16], b[16];
OUT out[16];
PARTS:
And(a=a[0], b=b[0], out=out[0]);
And(a=a[1], b=b[1], out=out[1]);
And(a=a[2], b=b[2], out=out[2]);
And(a=a[3], b=b[3], out=out[3]);
And(a=a[4], b=b[4], out=out[4]);
And(a=a[5], b=b[5], out=out[5]);
And(a=a[6], b=b[6], out=out[6]);
And(a=a[7], b=b[7], out=out[7]);
And(a=a[8], b=b[8], out=out[8]);
And(a=a[9], b=b[9], out=out[9]);
And(a=a[10], b=b[10], out=out[10]);
And(a=a[11], b=b[11], out=out[11]);
And(a=a[12], b=b[12], out=out[12]);
And(a=a[13], b=b[13], out=out[13]);
And(a=a[14], b=b[14], out=out[14]);
And(a=a[15], b=b[15], out=out[15]);
}
Chip And16
Eval Reset Clock: 0
Input pins
a 0 0 0 1 0 0 1 0 0 0 1 1 0 1 0 0 dec
b 1 0 0 1 1 0 0 0 0 1 1 1 0 1 1 0 dec
Output pins
out 0 0 0 1 0 0 0 0 0 0 1 1 0 1 0 0 dec
Test
Steps: 9 Outputs: 7
Slow Fast
Test Script
Compare File
Output File
Diff Table
set a %B0000000000000000,
set b %B0000000000000000,
eval,
output;
set a %B0000000000000000,
set b %B1111111111111111,
eval,
output;
set a %B1111111111111111,
set b %B1111111111111111,
eval,
output;
set a %B1010101010101010,
set b %B0101010101010101,
eval,
output;
set a %B0011110011000011,
set b %B0000111111110000,
eval,
output;
set a %B0001001000110100,
set b %B1001100001110110,
eval,
output;
Simulation successful: The output file is identical to the compare file

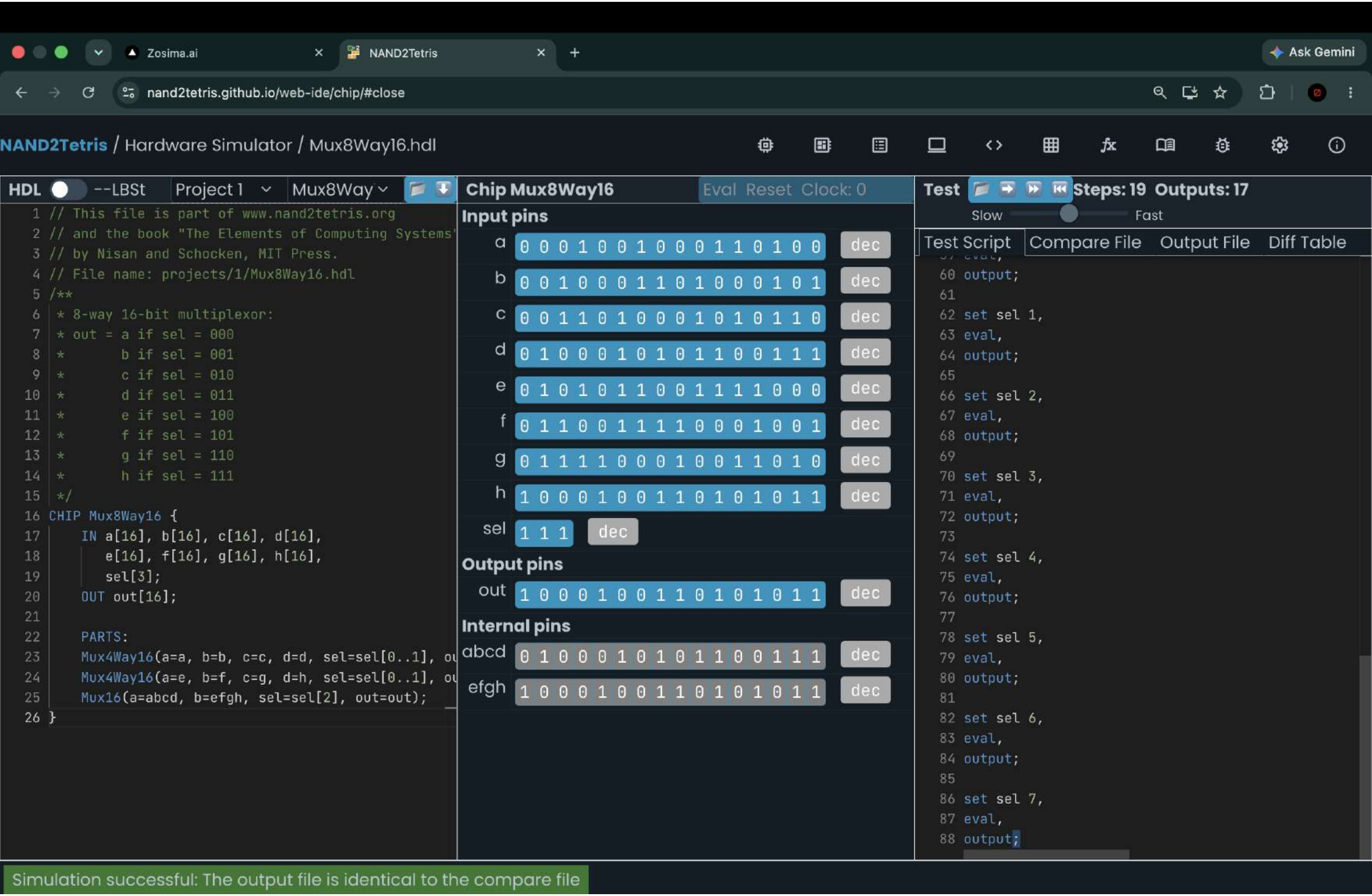

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NAND2Tetris
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NAND2Tetris / Hardware Simulator / DMux.hdl
HDL --LBSt
Project 1
DMux
// by Nisan and Schocken, MIT Press.
// File name: projects/1/DMux.hdl
/**
* Demultiplexor:
* [a, b] = [in, 0] if sel = 0
*   [0, in] if sel = 1
*/
CHIP DMux {
IN in, sel;
OUT a, b;
PARTS:
Not(in=sel, out=notSel);
And(a=in, b=notSel, out=a);
And(a=in, b=sel, out=b);
}
Chip DMux
Eval Reset Clock: 0
Input pins
in 1
sel 1
Output pins
a 0
b 1
Internal pins
notSel 0
Test
Steps: 7 Outputs: 5
Slow
Fast
Test Script
Compare File
Output File
Diff Table
set in 1,
set sel 0,
eval,
output;
set sel 1,
eval,
output;
Simulation successful: The output file is identical to the compare file

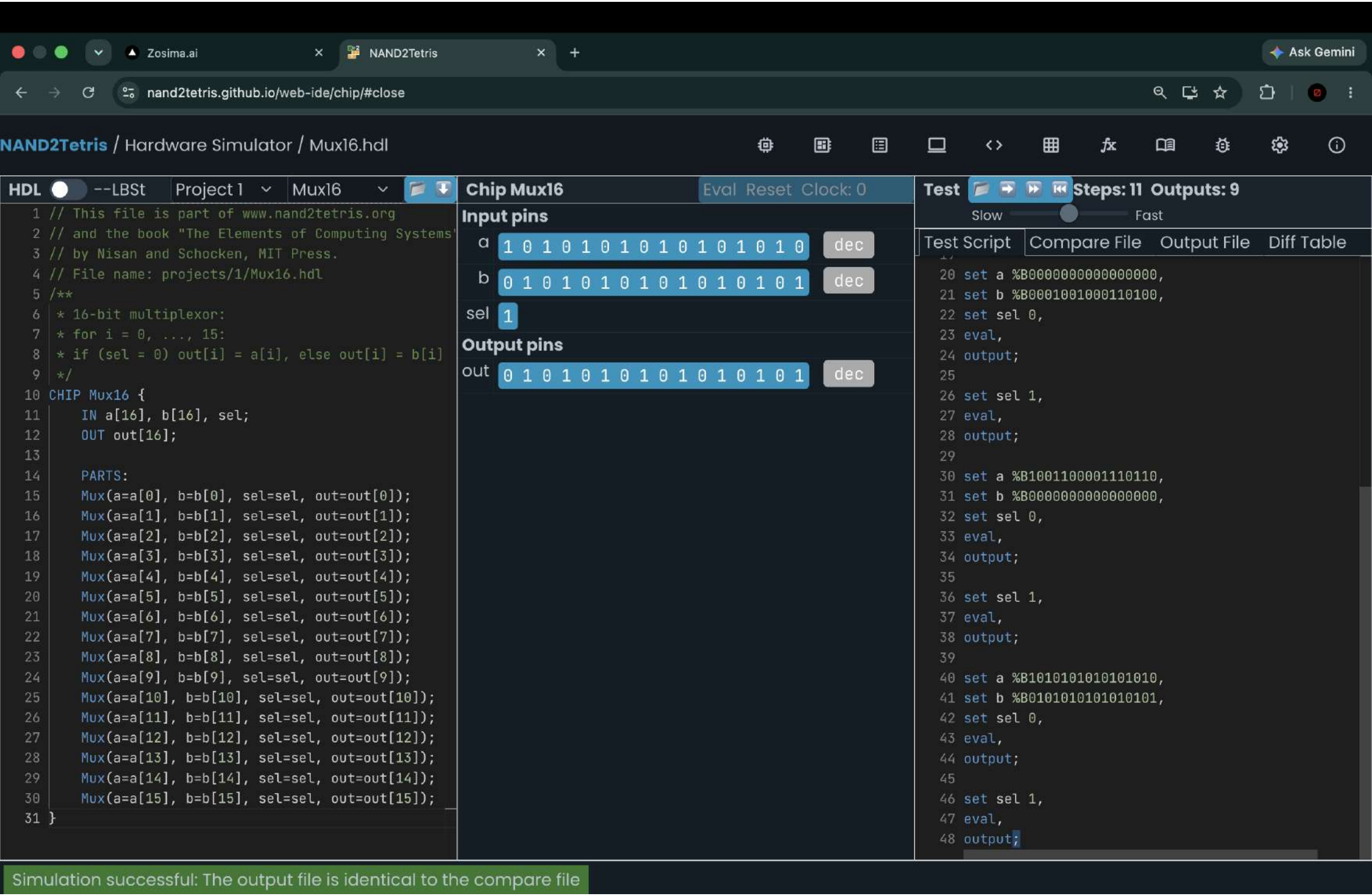

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NAND2Tetris / Hardware Simulator / DMux4Way.hdl
HDL --LBSt Project 1 DMux4Wa
// This file is part of www.nand2tetris.org
// and the book "The Elements of Computing Systems"
// by Nisan and Schocken, MIT Press.
// File name: projects/1/DMux4Way.hdl
/**
* 4-way demultiplexor:
* [a, b, c, d] = [in, 0, 0, 0] if sel = 00
* [0, in, 0, 0] if sel = 01
* [0, 0, in, 0] if sel = 10
* [0, 0, 0, in] if sel = 11
*/
CHIP DMux4Way {
IN in, sel[2];
OUT a, b, c, d;
PARTS:
DMux(in=in, sel=sel[1], a=top, b=bottom);
DMux(in=top, sel=sel[0], a=a, b=b);
DMux(in=bottom, sel=sel[0], a=c, b=d);
}
Chip DMux4Way
Eval Reset Clock: 0
Input pins
in 1
sel 1 1 dec
Output pins
a 0
b 0
c 0
d 1
Internal pins
top 0
bottom 1
Test
Steps: 11 Outputs: 9
Slow Fast
Test Script
Compare File
Output File
Diff Table
set sel %B01,
eval,
output;
set sel %B10,
eval,
output;
set sel %B11,
eval,
output;
set in 1,
set sel %B00,
eval,
output;
set sel %B01,
eval,
output;
set sel %B10,
eval,
output;
set sel %B11,
eval,
output;
Simulation successful: The output file is identical to the compare file

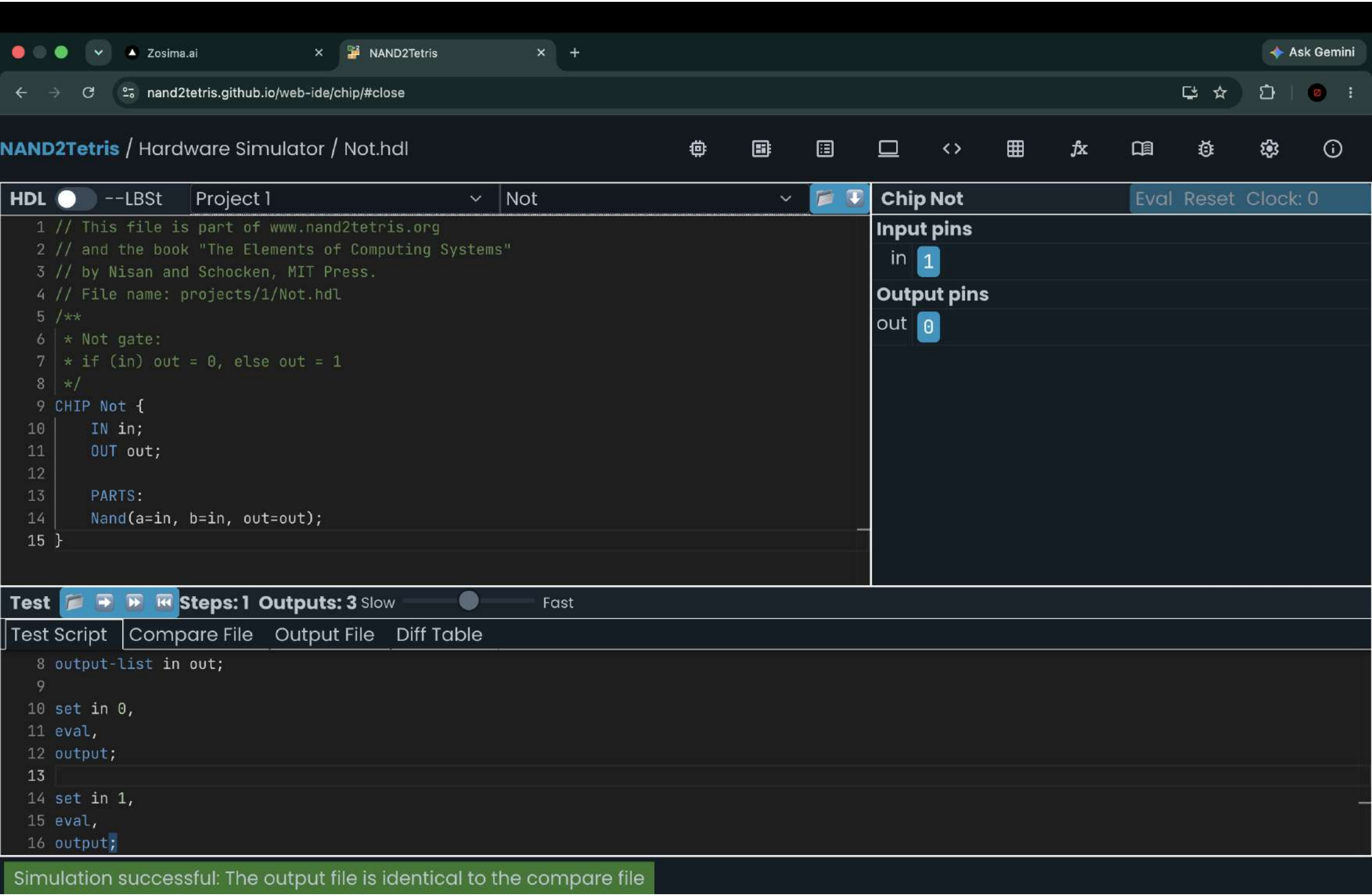

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NAND2Tetris / Hardware Simulator / DMux8Way.hdl
HDL --LBSt Project 1 DMux8Wa
// This file is part of www.nand2tetris.org
// and the book "The Elements of Computing Systems"
// by Nisan and Schocken, MIT Press.
// File name: projects/1/DMux8Way.hdl
/**
* 8-way demultiplexor:
* [a, b, c, d, e, f, g, h] = [in, 0, 0, 0, 0,
* [0, in, 0, 0, 0,
* [0, 0, in, 0, 0,
* [0, 0, 0, in, 0,
* [0, 0, 0, 0, in,
* [0, 0, 0, 0, 0,
* [0, 0, 0, 0, 0,
* [0, 0, 0, 0, 0,
*/
CHIP DMux8Way {
IN in, sel[3];
OUT a, b, c, d, e, f, g, h;
PARTS:
DMux4Way(in=in, sel=sel[1..2], a=ab, b=cd, c=ef, d=
DMux(in=ab, sel=sel[0], a=a, b=b);
DMux(in=cd, sel=sel[0], a=c, b=d);
DMux(in=ef, sel=sel[0], a=e, b=f);
DMux(in=gh, sel=sel[0], a=g, b=h);
}
Chip DMux8Way
Eval Reset Clock: 0
Input pins
in 1
sel 1 1 1 dec
Output pins
a 0
b 0
c 0
d 0
e 0
f 0
g 0
h 1
Internal pins
ab 0
cd 0
ef 0
gh 1
Test
Steps: 18 Outputs: 17
Slow Fast
Test Script
Compare File
Output File
Diff Table
output;
set sel %B001,
eval,
output;
set sel %B010,
eval,
output;
set sel %B011,
eval,
output;
set sel %B100,
eval,
output;
set sel %B101,
eval,
output;
set sel %B110,
eval,
output;
set sel %B111,
eval,
output;
Simulation successful: The output file is identical to the compare file

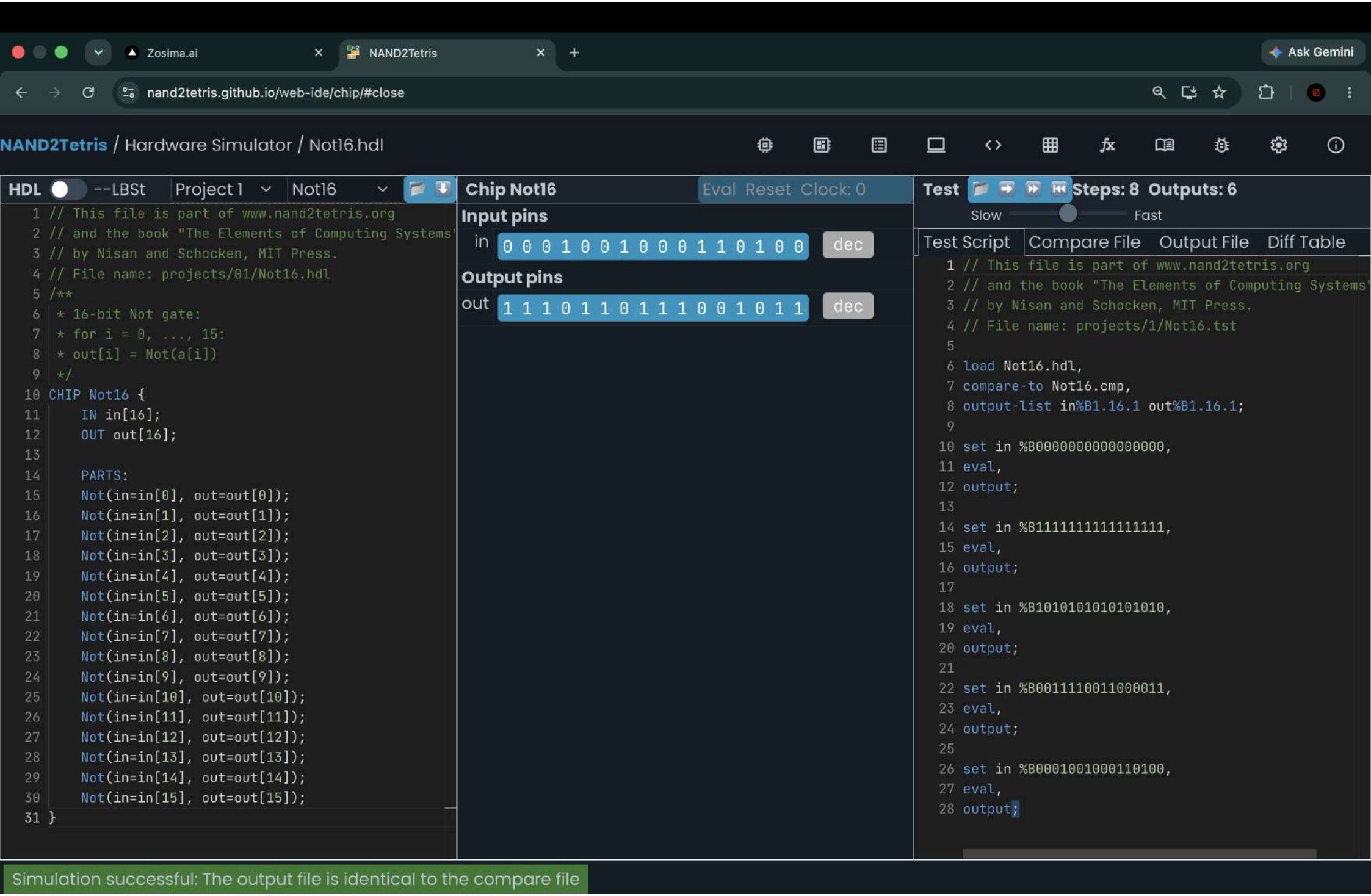

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NAND2Tetris / Hardware Simulator / Mux.hdl
HDL --LBSt
Project 1
Mux
// by Nisan and Schocken, MIT Press.
// File name: projects/1/Mux.hdl
/**
* Multiplexor:
* if (sel = 0) out = a, else out = b
*/
CHIP Mux {
IN a, b, sel;
OUT out;
PARTS:
Not(in=sel, out=notSel);
And(a=a, b=notSel, out=aSelected);
And(a=b, b=sel, out=bSelected);
Or(a=aSelected, b=bSelected, out=out);
}
Chip Mux
Eval Reset Clock: 0
Input pins
a 1
b 1
sel 1
Output pins
out 1
Internal pins
notSel 0
aSelected 0
bSelected 1
Test
Steps: 11 Outputs: 9 Slow
Fast
Test Script
Compare File
Output File
Diff Table
set a 1,
set b 1,
set sel 0,
eval,
output;
set sel 1,
eval,
output;
Simulation successful: The output file is identical to the compare file

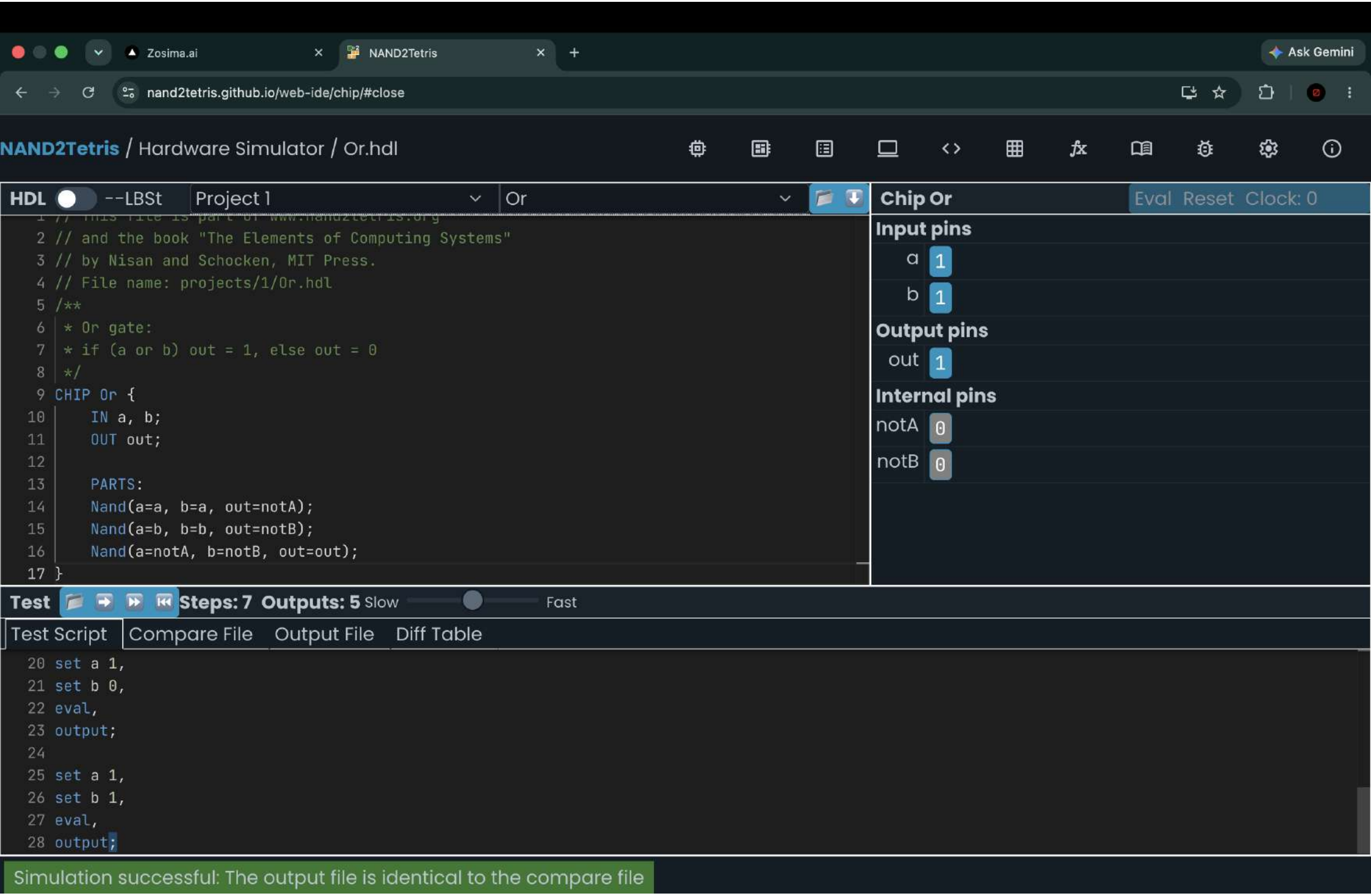

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NAND2Tetris / Hardware Simulator / Mux4Way16.hdl
HDL --LBSt Project 1 Mux4Way
// This file is part of www.nand2tetris.org
// and the book "The Elements of Computing Systems"
// by Nisan and Schocken, MIT Press.
// File name: projects/1/Mux4Way16.hdl
/**
* 4-way 16-bit multiplexor:
* out = a if sel = 00
* b if sel = 01
* c if sel = 10
* d if sel = 11
*/
CHIP Mux4Way16 {
IN a[16], b[16], c[16], d[16], sel[2];
OUT out[16];
PARTS:
Mux16(a=a, b=b, sel=sel[0], out=ab);
Mux16(a=c, b=d, sel=sel[0], out=cd);
Mux16(a=ab, b=cd, sel=sel[1], out=out);
}
Chip Mux4Way16
Eval Reset Clock: 0
Input pins
a 0 0 0 1 0 0 1 0 0 0 1 1 0 1 0 0 dec
b 1 0 0 1 1 0 0 0 0 1 1 1 0 1 1 0 dec
c 1 0 1 0 1 0 1 0 1 0 1 0 1 0 1 0 dec
d 0 1 0 1 0 1 0 1 0 1 0 1 0 1 0 1 dec
sel 1 1 dec
Output pins
out 0 1 0 1 0 1 0 1 0 1 0 1 0 1 0 1 dec
Internal pins
ab 1 0 0 1 1 0 0 0 0 1 1 1 0 1 1 0 dec
cd 0 1 0 1 0 1 0 1 0 1 0 1 0 1 0 1 dec
Test
Steps: 11 Outputs: 9
Slow
Fast
Test Script
Compare File
Output File
Diff Table
output;
set sel 2,
eval,
output;
set sel 3,
eval,
output;
set a %B0001001000110100,
set b %B1001100001110110,
set c %B1010101010101010,
set d %B0101010101010101,
set sel 0,
eval,
output;
set sel 1,
eval,
output;
set sel 2,
eval,
output;
set sel 3,
eval,
output;
Simulation successful: The output file is identical to the compare file

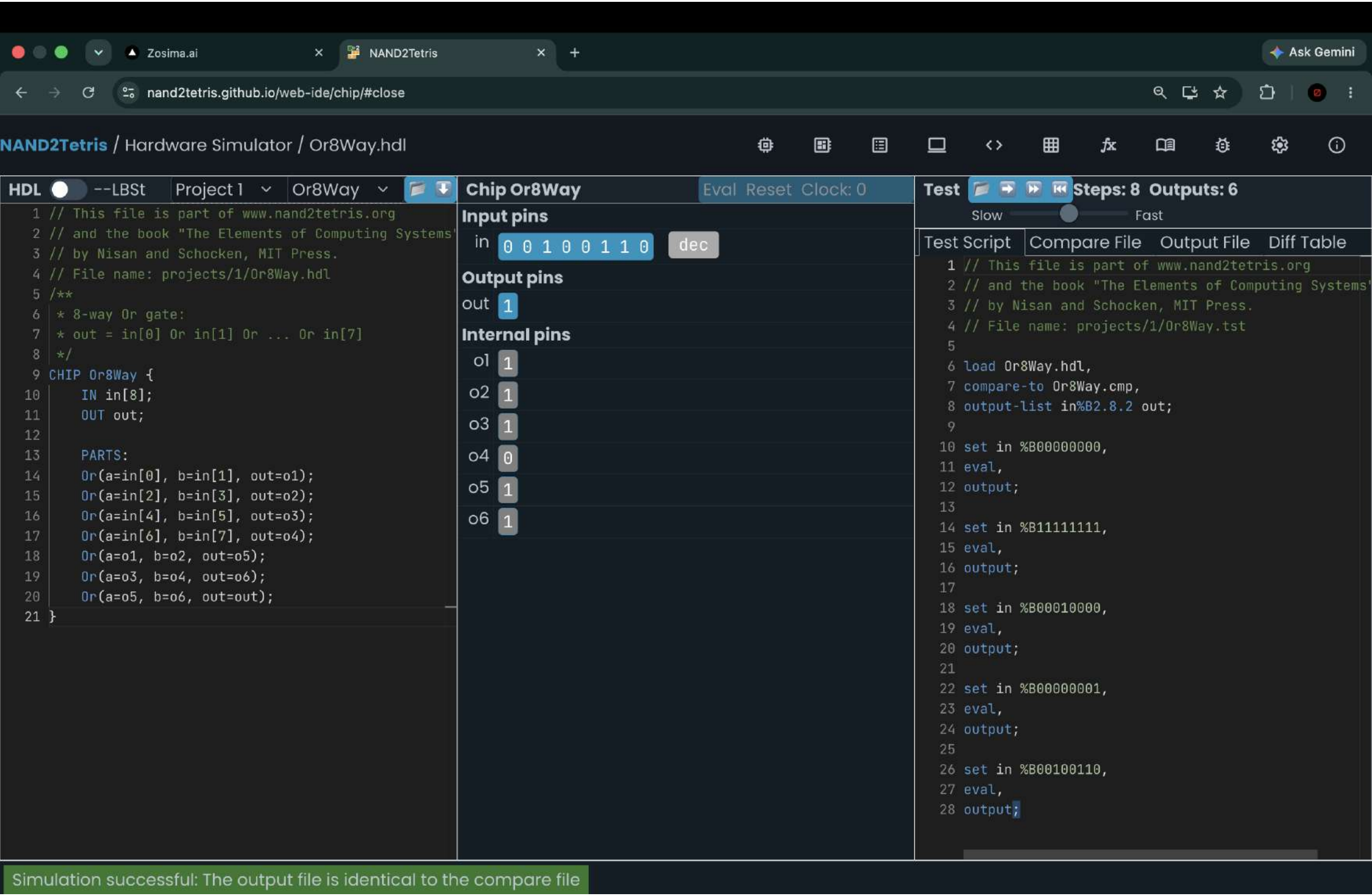

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NAND2Tetris / Hardware Simulator / Mux8Way16.hdl
HDL --LBSt Project 1 Mux8Way
// This file is part of www.nand2tetris.org
// and the book "The Elements of Computing Systems"
// by Nisan and Schocken, MIT Press.
// File name: projects/1/Mux8Way16.hdl
/**
* 8-way 16-bit multiplexor:
* out = a if sel = 000
* b if sel = 001
* c if sel = 010
* d if sel = 011
* e if sel = 100
* f if sel = 101
* g if sel = 110
* h if sel = 111
*/
CHIP Mux8Way16 {
IN a[16], b[16], c[16], d[16],
e[16], f[16], g[16], h[16],
sel[3];
OUT out[16];
PARTS:
Mux4Way16(a=a, b=b, c=c, d=d, sel=sel[0..1], ou
Mux4Way16(a=e, b=f, c=g, d=h, sel=sel[0..1], ou
Mux16(a=abcd, b=efgh, sel=sel[2], out=out);
}
Chip Mux8Way16
Eval Reset Clock: 0
Input pins
a 0 0 0 1 0 0 1 0 0 0 1 1 0 1 0 0 dec
b 0 0 1 0 0 0 1 1 0 1 0 0 0 1 0 1 dec
c 0 0 1 1 0 1 0 0 0 1 0 1 0 1 1 0 dec
d 0 1 0 0 0 1 0 1 0 1 1 0 0 1 1 1 dec
e 0 1 0 1 0 1 1 0 0 1 1 1 1 0 0 0 dec
f 0 1 1 0 0 1 1 1 1 0 0 0 1 0 0 1 dec
g 0 1 1 1 1 0 0 0 1 0 0 1 1 0 1 0 dec
h 1 0 0 0 1 0 0 1 1 0 1 0 1 0 1 1 dec
sel 1 1 1 dec
Output pins
out 1 0 0 0 1 0 0 1 1 0 1 0 1 0 1 1 dec
Internal pins
abcd 0 1 0 0 0 1 0 1 0 1 1 0 0 1 1 1 dec
efgh 1 0 0 0 1 0 0 1 1 0 1 0 1 0 1 1 dec
Test
Steps: 19 Outputs: 17
Slow Fast
Test Script
Compare File
Output File
Diff Table
output;
set sel 1,
eval,
output;
set sel 2,
eval,
output;
set sel 3,
eval,
output;
set sel 4,
eval,
output;
set sel 5,
eval,
output;
set sel 6,
eval,
output;
set sel 7,
eval,
output;
Simulation successful: The output file is identical to the compare file

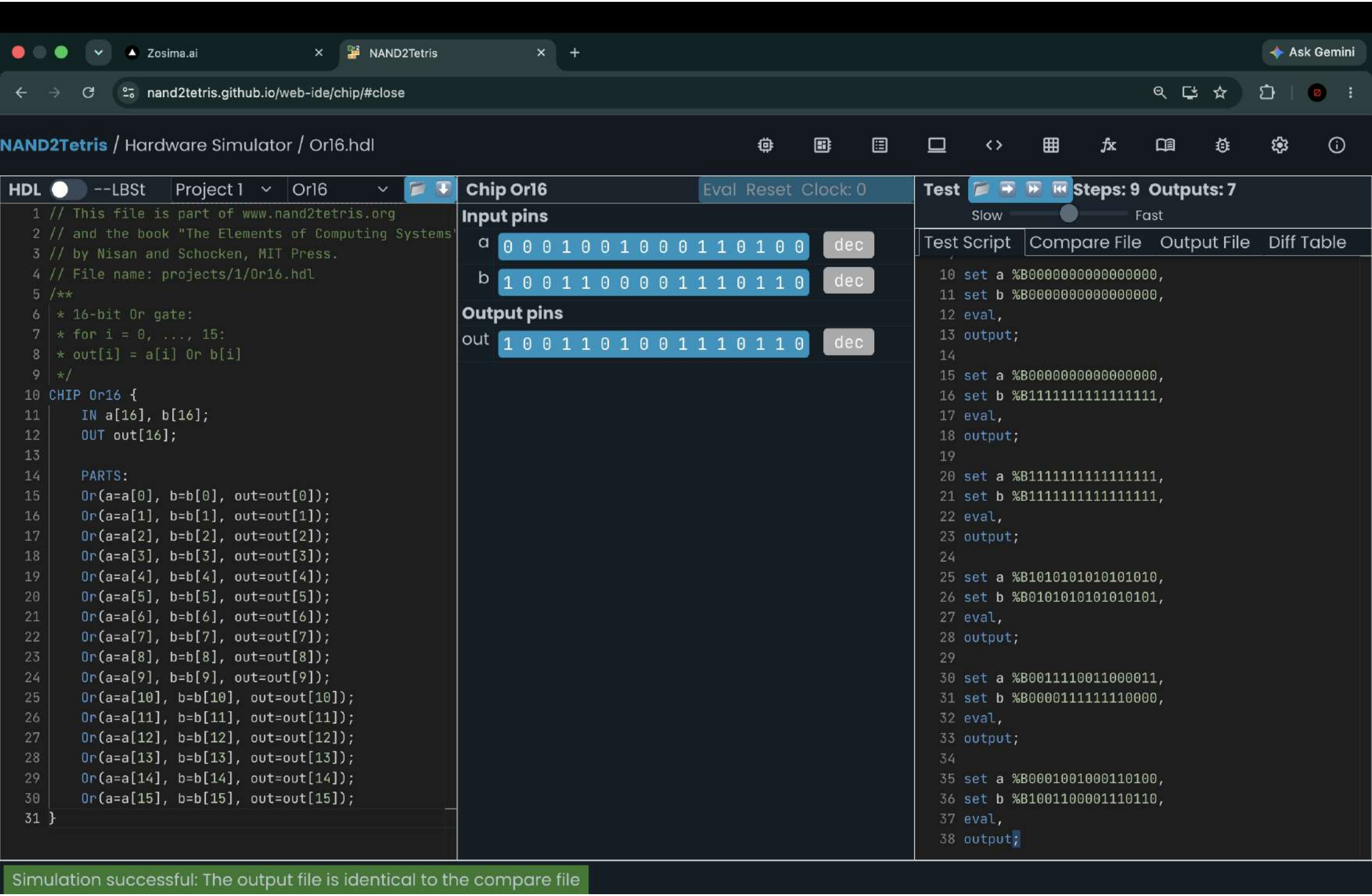

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nand2tetris.github.io/web-ide/chip/#close
NAND2Tetris / Hardware Simulator / Mux16.hdl
HDL --LBSt Project 1 Mux16
// This file is part of www.nand2tetris.org
// and the book "The Elements of Computing Systems"
// by Nisan and Schocken, MIT Press.
// File name: projects/1/Mux16.hdl
/**
* 16-bit multiplexor:
* for i = 0, ..., 15:
* if (sel = 0) out[i] = a[i], else out[i] = b[i]
*/
CHIP Mux16 {
IN a[16], b[16], sel;
OUT out[16];
PARTS:
Mux(a=a[0], b=b[0], sel=sel, out=out[0]);
Mux(a=a[1], b=b[1], sel=sel, out=out[1]);
Mux(a=a[2], b=b[2], sel=sel, out=out[2]);
Mux(a=a[3], b=b[3], sel=sel, out=out[3]);
Mux(a=a[4], b=b[4], sel=sel, out=out[4]);
Mux(a=a[5], b=b[5], sel=sel, out=out[5]);
Mux(a=a[6], b=b[6], sel=sel, out=out[6]);
Mux(a=a[7], b=b[7], sel=sel, out=out[7]);
Mux(a=a[8], b=b[8], sel=sel, out=out[8]);
Mux(a=a[9], b=b[9], sel=sel, out=out[9]);
Mux(a=a[10], b=b[10], sel=sel, out=out[10]);
Mux(a=a[11], b=b[11], sel=sel, out=out[11]);
Mux(a=a[12], b=b[12], sel=sel, out=out[12]);
Mux(a=a[13], b=b[13], sel=sel, out=out[13]);
Mux(a=a[14], b=b[14], sel=sel, out=out[14]);
Mux(a=a[15], b=b[15], sel=sel, out=out[15]);
}
Chip Mux16
Eval Reset Clock: 0
Input pins
a 1 0 1 0 1 0 1 0 1 0 1 0 1 0 1 0 dec
b 0 1 0 1 0 1 0 1 0 1 0 1 0 1 0 1 dec
sel 1
Output pins
out 0 1 0 1 0 1 0 1 0 1 0 1 0 1 0 1 dec
Test
Steps: 11 Outputs: 9
Slow Fast
Test Script
Compare File
Output File
Diff Table
set a %B0000000000000000,
set b %B0001001000110100,
set sel 0,
eval,
output;
set sel 1,
eval,
output;
set a %B1001100001110110,
set b %B0000000000000000,
set sel 0,
eval,
output;
set sel 1,
eval,
output;
set a %B1010101010101010,
set b %B0101010101010101,
set sel 0,
eval,
output;
set sel 1,
eval,
output;
Simulation successful: The output file is identical to the compare file

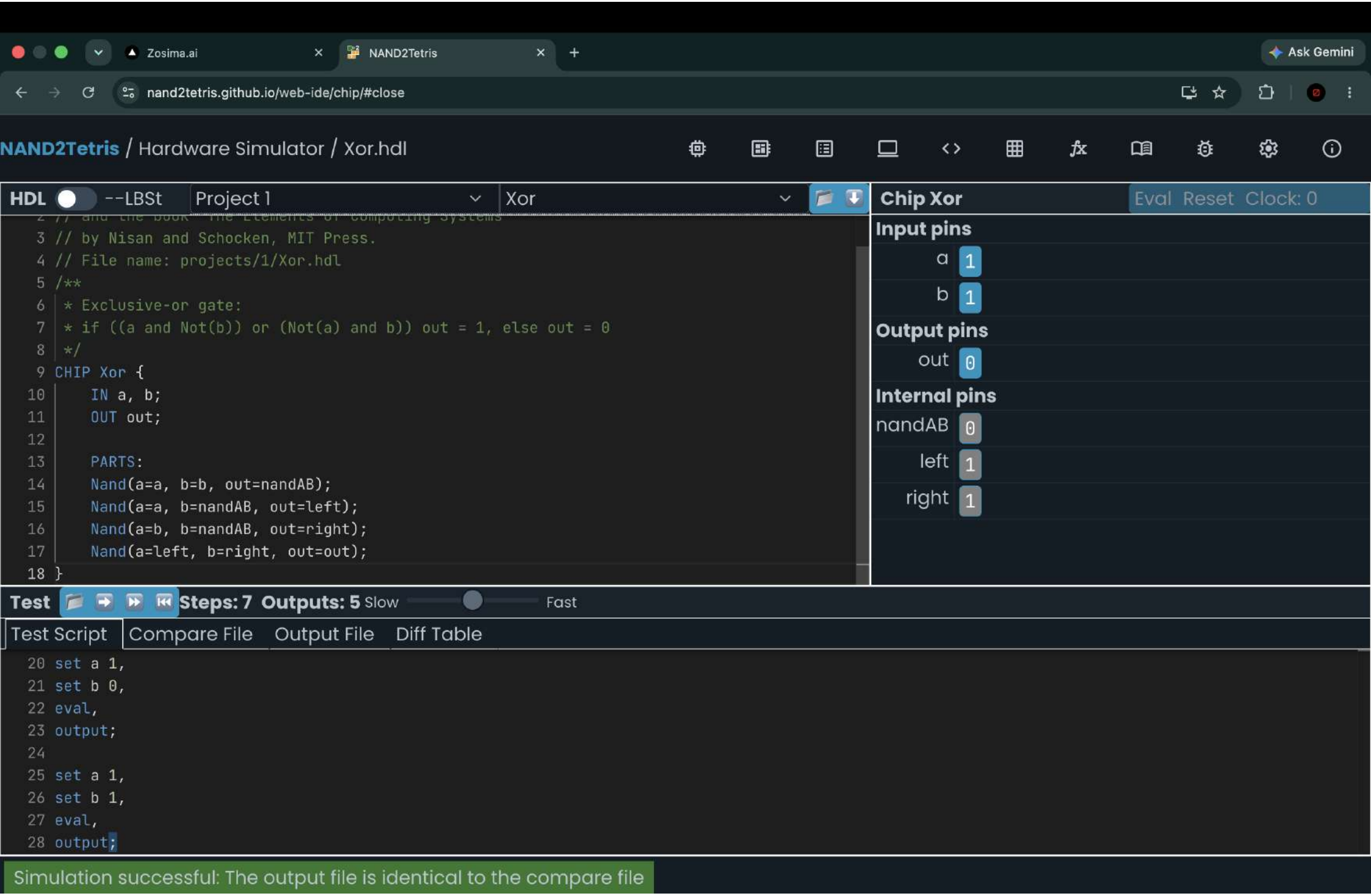

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NAND2Tetris
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NAND2Tetris / Hardware Simulator / Not.hdl
HDL --LBSt
Project 1
Not
1 // This file is part of www.nand2tetris.org
2 // and the book "The Elements of Computing Systems"
3 // by Nisan and Schocken, MIT Press.
4 // File name: projects/1/Not.hdl
5 /**
6 * Not gate:
7 * if (in) out = 0, else out = 1
8 */
9 CHIP Not {
10 IN in;
11 OUT out;
12
13 PARTS:
14 Nand(a=in, b=in, out=out);
15 }
Chip Not
Eval Reset Clock: 0
Input pins
in 1
Output pins
out 0
Test
Steps: 1 Outputs: 3
Slow
Fast
Test Script
Compare File
Output File
Diff Table
8 output-list in out;
9
10 set in 0,
11 eval,
12 output;
13
14 set in 1,
15 eval,
16 output;
Simulation successful: The output file is identical to the compare file

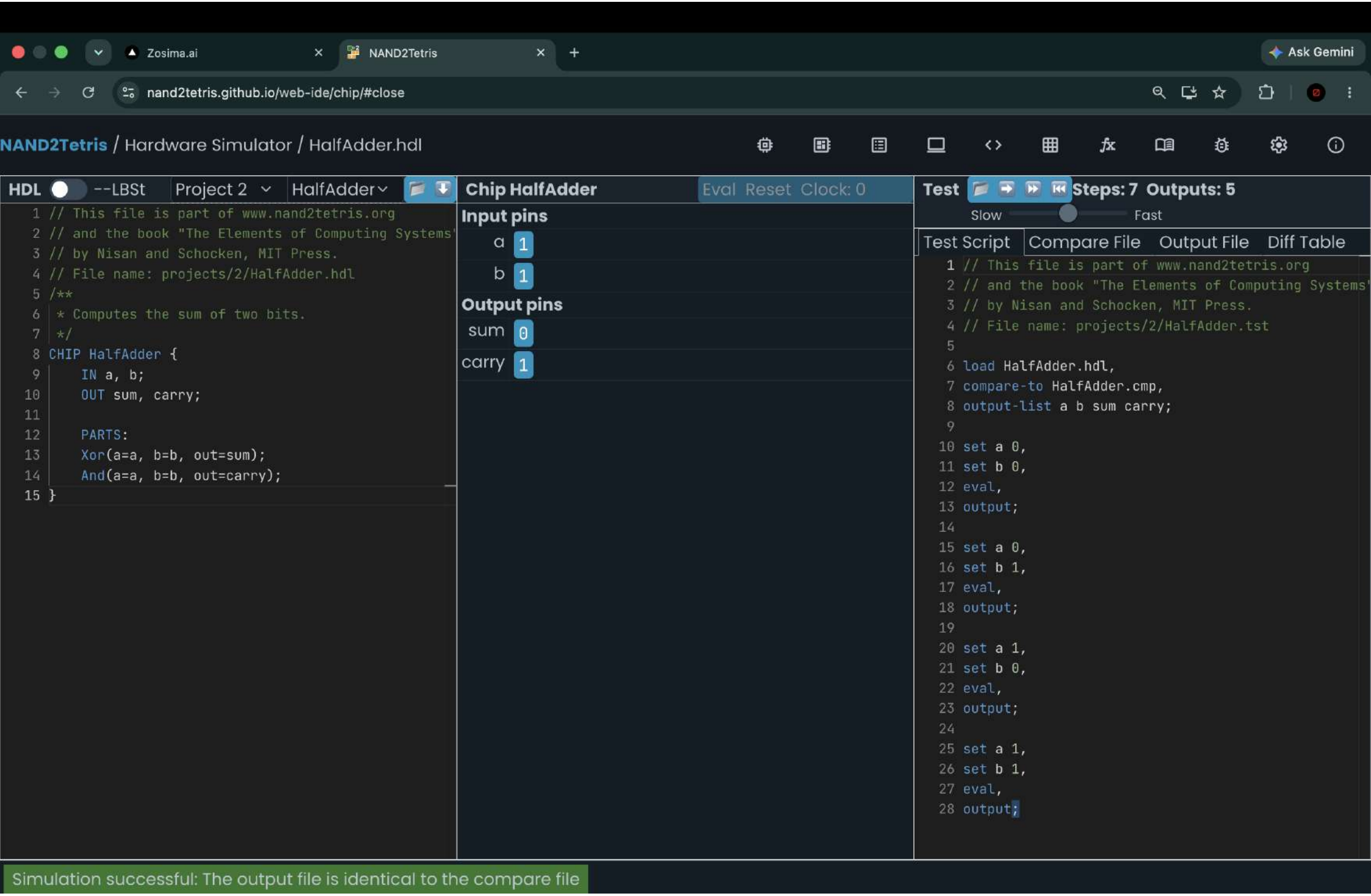

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NAND2Tetris
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nand2tetris.github.io/web-ide/chip/#close
NAND2Tetris / Hardware Simulator / Not16.hdl
HDL --LBSt Project 1 Not16
// This file is part of www.nand2tetris.org
// and the book "The Elements of Computing Systems"
// by Nisan and Schocken, MIT Press.
// File name: projects/01/Not16.hdl
/**
* 16-bit Not gate:
* for i = 0, ..., 15:
* out[i] = Not(a[i])
*/
CHIP Not16 {
IN in[16];
OUT out[16];
PARTS:
Not(in=in[0], out=out[0]);
Not(in=in[1], out=out[1]);
Not(in=in[2], out=out[2]);
Not(in=in[3], out=out[3]);
Not(in=in[4], out=out[4]);
Not(in=in[5], out=out[5]);
Not(in=in[6], out=out[6]);
Not(in=in[7], out=out[7]);
Not(in=in[8], out=out[8]);
Not(in=in[9], out=out[9]);
Not(in=in[10], out=out[10]);
Not(in=in[11], out=out[11]);
Not(in=in[12], out=out[12]);
Not(in=in[13], out=out[13]);
Not(in=in[14], out=out[14]);
Not(in=in[15], out=out[15]);
}
Chip Not16
Eval Reset Clock: 0
Input pins
in 0 0 0 1 0 0 1 0 0 0 1 1 0 1 0 0
dec
Output pins
out 1 1 1 0 1 1 0 1 1 1 0 0 1 0 1 1
dec
Test
Steps: 8 Outputs: 6
Slow
Fast
Test Script
Compare File
Output File
Diff Table
// This file is part of www.nand2tetris.org
// and the book "The Elements of Computing Systems"
// by Nisan and Schocken, MIT Press.
// File name: projects/1/Not16.tst
load Not16.hdl,
compare-to Not16.cmp,
output-list in%B1.16.1 out%B1.16.1;
set in %B0000000000000000,
eval,
output;
set in %B1111111111111111,
eval,
output;
set in %B1010101010101010,
eval,
output;
set in %B0011110011000011,
eval,
output;
set in %B0001001000110100,
eval,
output;
Simulation successful: The output file is identical to the compare file

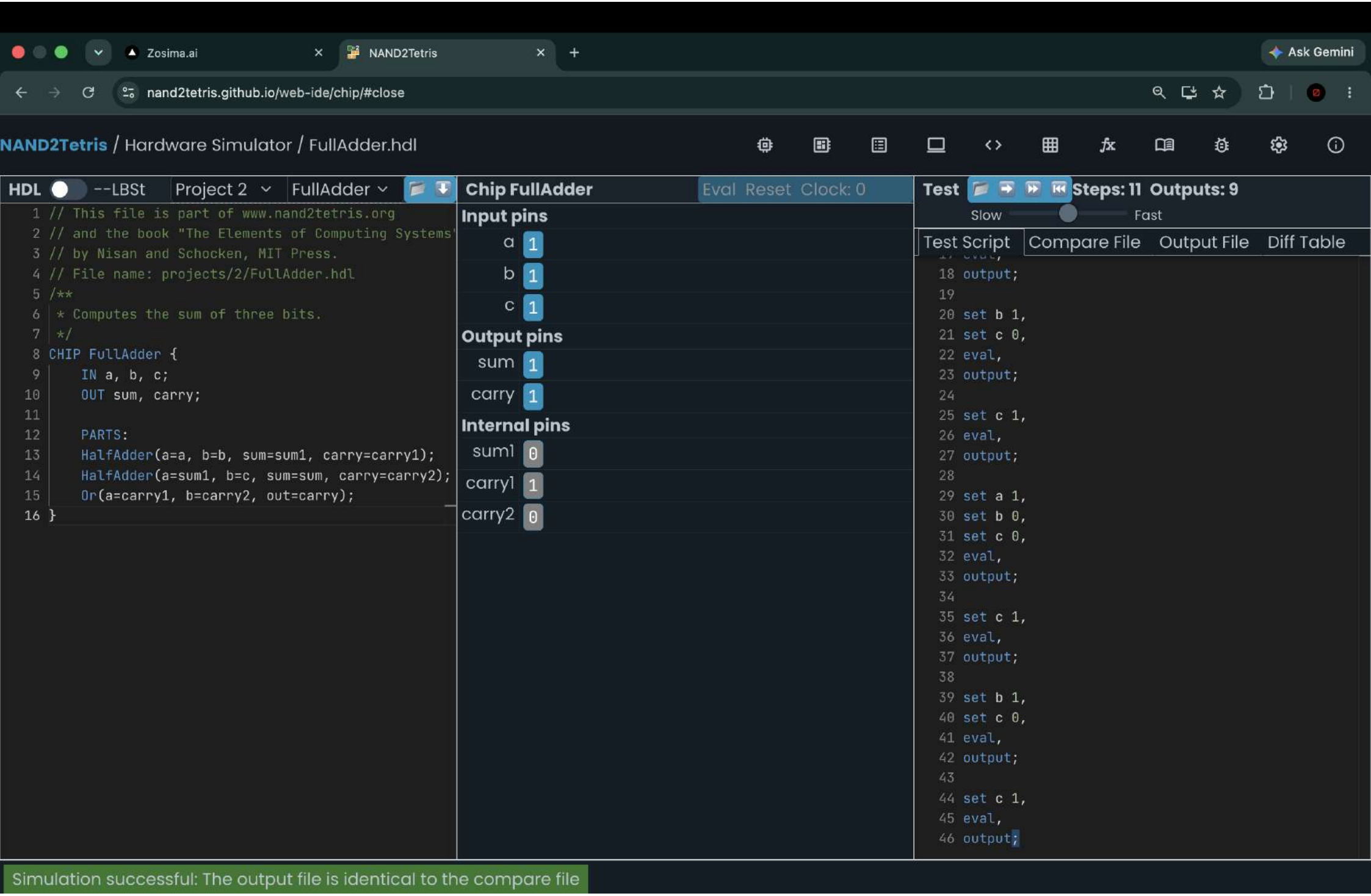

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NAND2Tetris / Hardware Simulator / Or.hdl
HDL --LBSt
Project 1
Or
// and the book "The Elements of Computing Systems"
// by Nisan and Schocken, MIT Press.
// File name: projects/1/Or.hdl
/**
* Or gate:
* if (a or b) out = 1, else out = 0
*/
CHIP Or {
IN a, b;
OUT out;
PARTS:
Nand(a=a, b=a, out=notA);
Nand(a=b, b=b, out=notB);
Nand(a=notA, b=notB, out=out);
}
Chip Or
Eval Reset Clock: 0
Input pins
a 1
b 1
Output pins
out 1
Internal pins
notA 0
notB 0
Test
Steps: 7 Outputs: 5
Slow
Fast
Test Script
Compare File
Output File
Diff Table
set a 1,
set b 0,
eval,
output;
set a 1,
set b 1,
eval,
output;
Simulation successful: The output file is identical to the compare file

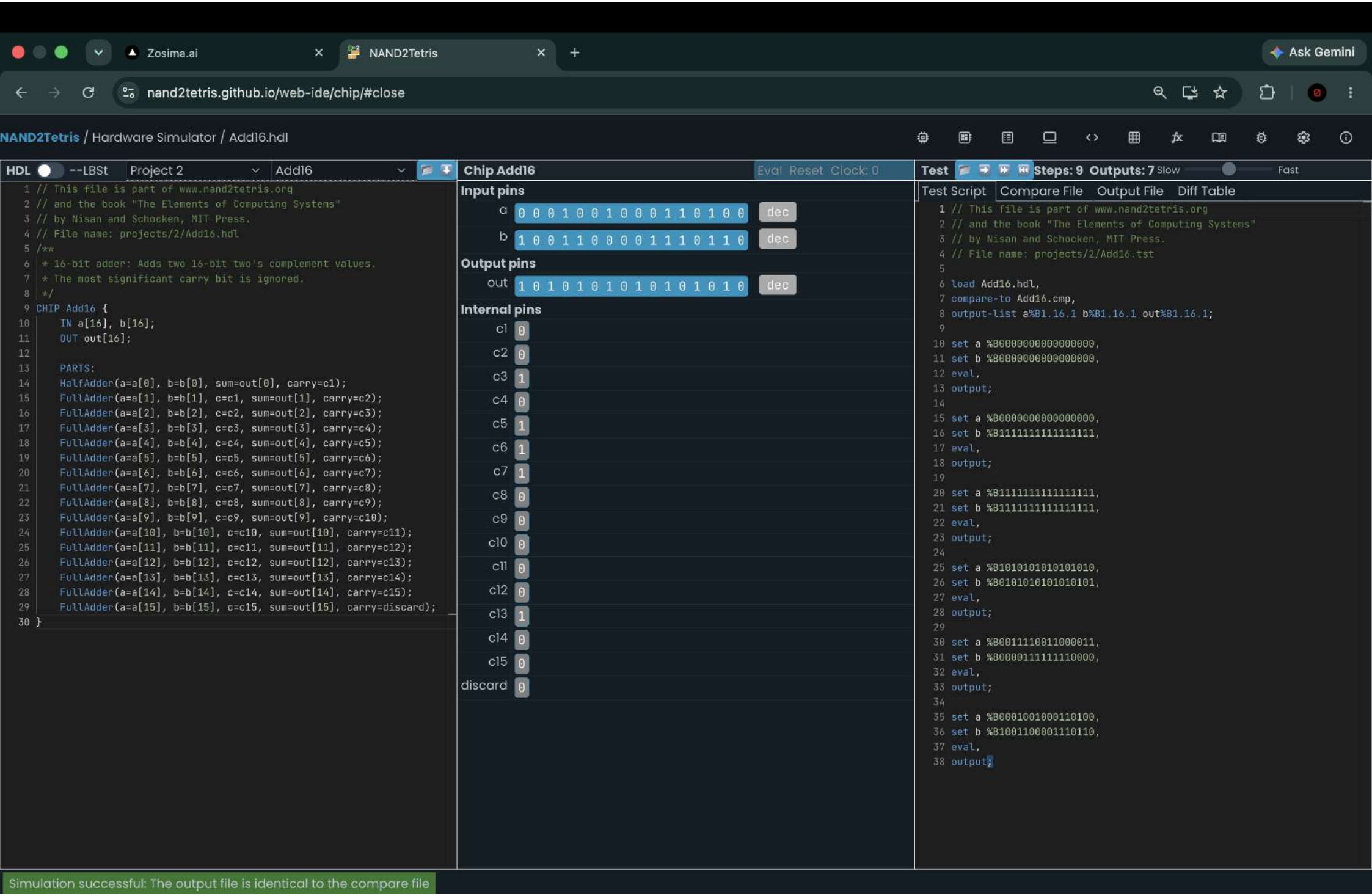

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NAND2Tetris / Hardware Simulator / Or8Way.hdl
HDL --LBSt Project 1 Or8Way
// This file is part of www.nand2tetris.org
// and the book "The Elements of Computing Systems"
// by Nisan and Schocken, MIT Press.
// File name: projects/1/Or8Way.hdl
/**
* 8-way Or gate:
* out = in[0] Or in[1] Or ... Or in[7]
*/
CHIP Or8Way {
IN in[8];
OUT out;
PARTS:
Or(a=in[0], b=in[1], out=o1);
Or(a=in[2], b=in[3], out=o2);
Or(a=in[4], b=in[5], out=o3);
Or(a=in[6], b=in[7], out=o4);
Or(a=o1, b=o2, out=o5);
Or(a=o3, b=o4, out=o6);
Or(a=o5, b=o6, out=out);
}
Chip Or8Way
Eval Reset Clock: 0
Input pins
in 0 0 1 0 0 1 1 0
dec
Output pins
out 1
Internal pins
o1 1
o2 1
o3 1
o4 0
o5 1
o6 1
Test
Steps: 8 Outputs: 6
Slow
Fast
Test Script
Compare File
Output File
Diff Table
// This file is part of www.nand2tetris.org
// and the book "The Elements of Computing Systems"
// by Nisan and Schocken, MIT Press.
// File name: projects/1/Or8Way.tst
load Or8Way.hdl,
compare-to Or8Way.cmp,
output-list in%B2.8.2 out;
set in %B00000000,
eval,
output;
set in %B11111111,
eval,
output;
set in %B00010000,
eval,
output;
set in %B00000001,
eval,
output;
set in %B00100110,
eval,
output;
Simulation successful: The output file is identical to the compare file

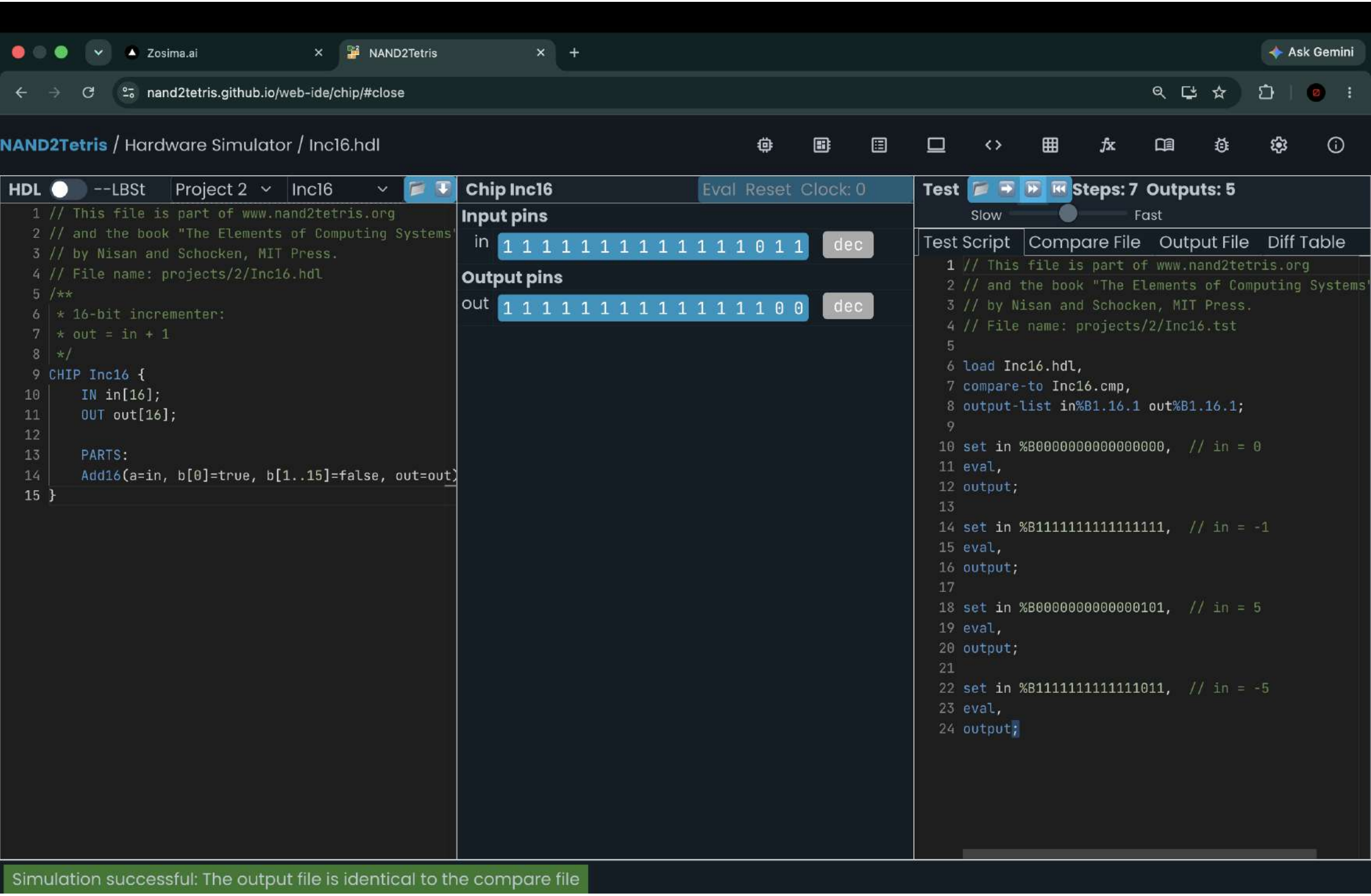

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NAND2Tetris / Hardware Simulator / Or16.hdl
HDL --LBSt Project 1 Or16
// This file is part of www.nand2tetris.org
// and the book "The Elements of Computing Systems"
// by Nisan and Schocken, MIT Press.
// File name: projects/1/Or16.hdl
/**
* 16-bit Or gate:
* for i = 0, ..., 15:
* out[i] = a[i] Or b[i]
*/
CHIP Or16 {
IN a[16], b[16];
OUT out[16];
PARTS:
Or(a=a[0], b=b[0], out=out[0]);
Or(a=a[1], b=b[1], out=out[1]);
Or(a=a[2], b=b[2], out=out[2]);
Or(a=a[3], b=b[3], out=out[3]);
Or(a=a[4], b=b[4], out=out[4]);
Or(a=a[5], b=b[5], out=out[5]);
Or(a=a[6], b=b[6], out=out[6]);
Or(a=a[7], b=b[7], out=out[7]);
Or(a=a[8], b=b[8], out=out[8]);
Or(a=a[9], b=b[9], out=out[9]);
Or(a=a[10], b=b[10], out=out[10]);
Or(a=a[11], b=b[11], out=out[11]);
Or(a=a[12], b=b[12], out=out[12]);
Or(a=a[13], b=b[13], out=out[13]);
Or(a=a[14], b=b[14], out=out[14]);
Or(a=a[15], b=b[15], out=out[15]);
}
Chip Or16 Eval Reset Clock: 0
Input pins
a 0 0 0 1 0 0 1 0 0 0 1 1 0 1 0 0 dec
b 1 0 0 1 1 0 0 0 0 1 1 1 0 1 1 0 dec
Output pins
out 1 0 0 1 1 0 1 0 0 1 1 1 0 1 1 0 dec
Test Steps: 9 Outputs: 7
Slow Fast
Test Script Compare File Output File Diff Table
set a %B0000000000000000,
set b %B0000000000000000,
eval,
output;
set a %B0000000000000000,
set b %B1111111111111111,
eval,
output;
set a %B1111111111111111,
set b %B1111111111111111,
eval,
output;
set a %B1010101010101010,
set b %B0101010101010101,
eval,
output;
set a %B0011110011000011,
set b %B0000111111110000,
eval,
output;
set a %B0001001000110100,
set b %B1001100001110110,
eval,
output;
Simulation successful: The output file is identical to the compare file

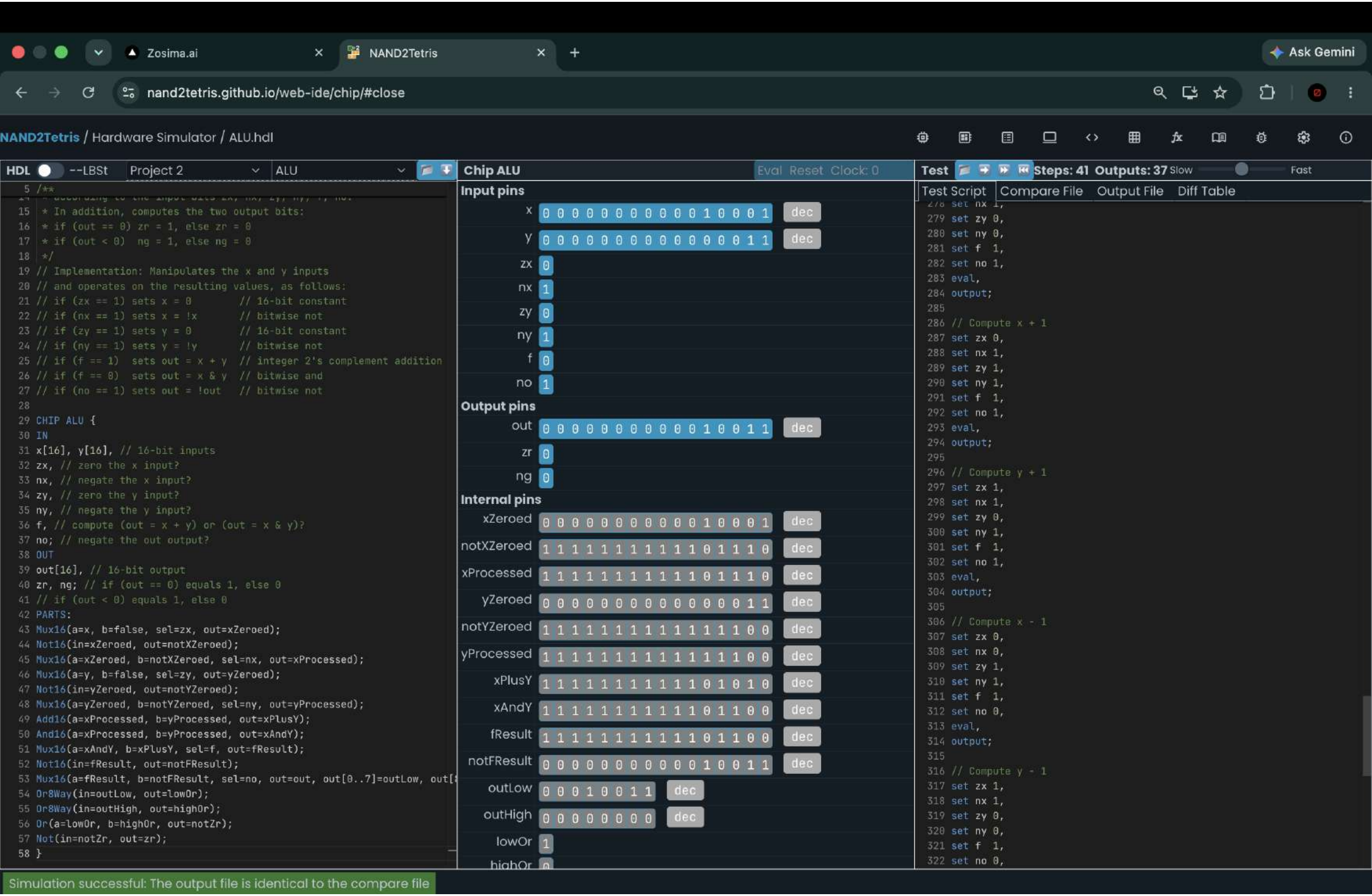

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NAND2Tetris / Hardware Simulator / Xor.hdl
HDL --LBSt
Project 1
Xor
// by Nisan and Schocken, MIT Press.
// File name: projects/1/Xor.hdl
/**
* Exclusive-or gate:
* if ((a and Not(b)) or (Not(a) and b)) out = 1, else out = 0
*/
CHIP Xor {
IN a, b;
OUT out;
PARTS:
Nand(a=a, b=b, out=nandAB);
Nand(a=a, b=nandAB, out=left);
Nand(a=b, b=nandAB, out=right);
Nand(a=left, b=right, out=out);
}
Chip Xor
Eval Reset Clock: 0
Input pins
a 1
b 1
Output pins
out 0
Internal pins
nandAB 0
left 1
right 1
Test
Steps: 7 Outputs: 5
Slow
Fast
Test Script
Compare File
Output File
Diff Table
set a 1,
set b 0,
eval,
output;
set a 1,
set b 1,
eval,
output;
Simulation successful: The output file is identical to the compare file


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NAND2Tetris / Hardware Simulator / HalfAdder.hdl
HDL --LBSt Project 2 HalfAdder
// This file is part of www.nand2tetris.org
// and the book "The Elements of Computing Systems"
// by Nisan and Schocken, MIT Press.
// File name: projects/2/HalfAdder.hdl
/**
* Computes the sum of two bits.
*/
CHIP HalfAdder {
IN a, b;
OUT sum, carry;
PARTS:
Xor(a=a, b=b, out=sum);
And(a=a, b=b, out=carry);
}
Chip HalfAdder Eval Reset Clock: 0
Input pins
a 1
b 1
Output pins
sum 0
carry 1
Test Steps: 7 Outputs: 5
Slow Fast
Test Script Compare File Output File Diff Table
// This file is part of www.nand2tetris.org
// and the book "The Elements of Computing Systems"
// by Nisan and Schocken, MIT Press.
// File name: projects/2/HalfAdder.tst
load HalfAdder.hdl,
compare-to HalfAdder.cmp,
output-list a b sum carry;
set a 0,
set b 0,
eval,
output;
set a 0,
set b 1,
eval,
output;
set a 1,
set b 0,
eval,
output;
set a 1,
set b 1,
eval,
output;
Simulation successful: The output file is identical to the compare file


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NAND2Tetris / Hardware Simulator / FullAdder.hdl
HDL --LBSt Project 2 FullAdder
1 // This file is part of www.nand2tetris.org
2 // and the book "The Elements of Computing Systems"
3 // by Nisan and Schocken, MIT Press.
4 // File name: projects/2/FullAdder.hdl
5 /**
6 * Computes the sum of three bits.
7 */
8 CHIP FullAdder {
9 IN a, b, c;
10 OUT sum, carry;
11
12 PARTS:
13 HalfAdder(a=a, b=b, sum=sum1, carry=carry1);
14 HalfAdder(a=sum1, b=c, sum=sum, carry=carry2);
15 Or(a=carry1, b=carry2, out=carry);
16 }
Chip FullAdder Eval Reset Clock: 0
Input pins
a 1
b 1
c 1
Output pins
sum 1
carry 1
Internal pins
sum1 0
carry1 1
carry2 0
Test Steps: 11 Outputs: 9
Slow Fast
Test Script Compare File Output File Diff Table
18 output;
19
20 set b 1,
21 set c 0,
22 eval,
23 output;
24
25 set c 1,
26 eval,
27 output;
28
29 set a 1,
30 set b 0,
31 set c 0,
32 eval,
33 output;
34
35 set c 1,
36 eval,
37 output;
38
39 set b 1,
40 set c 0,
41 eval,
42 output;
43
44 set c 1,
45 eval,
46 output;
Simulation successful: The output file is identical to the compare file


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NAND2Tetris / Hardware Simulator / Add16.hdl
HDL --LBSt Project 2 Add16
1 // This file is part of www.nand2tetris.org
2 // and the book "The Elements of Computing Systems"
3 // by Nisan and Schocken, MIT Press.
4 // File name: projects/2/Add16.hdl
5 /**
6 * 16-bit adder: Adds two 16-bit two's complement values.
7 * The most significant carry bit is ignored.
8 */
9 CHIP Add16 {
10 IN a[16], b[16];
11 OUT out[16];
12
13 PARTS:
14 HalfAdder(a=a[0], b=b[0], sum=out[0], carry=c1);
15 FullAdder(a=a[1], b=b[1], c=c1, sum=out[1], carry=c2);
16 FullAdder(a=a[2], b=b[2], c=c2, sum=out[2], carry=c3);
17 FullAdder(a=a[3], b=b[3], c=c3, sum=out[3], carry=c4);
18 FullAdder(a=a[4], b=b[4], c=c4, sum=out[4], carry=c5);
19 FullAdder(a=a[5], b=b[5], c=c5, sum=out[5], carry=c6);
20 FullAdder(a=a[6], b=b[6], c=c6, sum=out[6], carry=c7);
21 FullAdder(a=a[7], b=b[7], c=c7, sum=out[7], carry=c8);
22 FullAdder(a=a[8], b=b[8], c=c8, sum=out[8], carry=c9);
23 FullAdder(a=a[9], b=b[9], c=c9, sum=out[9], carry=c10);
24 FullAdder(a=a[10], b=b[10], c=c10, sum=out[10], carry=c11);
25 FullAdder(a=a[11], b=b[11], c=c11, sum=out[11], carry=c12);
26 FullAdder(a=a[12], b=b[12], c=c12, sum=out[12], carry=c13);
27 FullAdder(a=a[13], b=b[13], c=c13, sum=out[13], carry=c14);
28 FullAdder(a=a[14], b=b[14], c=c14, sum=out[14], carry=c15);
29 FullAdder(a=a[15], b=b[15], c=c15, sum=out[15], carry=discard);
30 }
Chip Add16
Eval Reset Clock: 0
Input pins
a 0 0 0 1 0 0 1 0 0 0 1 1 0 1 0 0 dec
b 1 0 0 1 1 0 0 0 0 1 1 1 0 1 1 0 dec
Output pins
out 1 0 1 0 1 0 1 0 1 0 1 0 1 0 1 0 dec
Internal pins
c1 0
c2 0
c3 1
c4 0
c5 1
c6 1
c7 1
c8 0
c9 0
c10 0
c11 0
c12 0
c13 1
c14 0
c15 0
discard 0
Test
Steps: 9 Outputs: 7 Slow Fast
Test Script
Compare File
Output File
Diff Table
1 // This file is part of www.nand2tetris.org
2 // and the book "The Elements of Computing Systems"
3 // by Nisan and Schocken, MIT Press.
4 // File name: projects/2/Add16.tst
5
6 load Add16.hdl,
7 compare-to Add16.cmp,
8 output-list a%B1.16.1 b%B1.16.1 out%B1.16.1;
9
10 set a %B0000000000000000,
11 set b %B0000000000000000,
12 eval,
13 output;
14
15 set a %B0000000000000000,
16 set b %B1111111111111111,
17 eval,
18 output;
19
20 set a %B1111111111111111,
21 set b %B1111111111111111,
22 eval,
23 output;
24
25 set a %B1010101010101010,
26 set b %B0101010101010101,
27 eval,
28 output;
29
30 set a %B0011110011000011,
31 set b %B0000111111110000,
32 eval,
33 output;
34
35 set a %B0001001000110100,
36 set b %B1001100001110110,
37 eval,
38 output;
Simulation successful: The output file is identical to the compare file


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NAND2Tetris / Hardware Simulator / Inc16.hdl
HDL --LBSt Project 2 Inc16
// This file is part of www.nand2tetris.org
// and the book "The Elements of Computing Systems"
// by Nisan and Schocken, MIT Press.
// File name: projects/2/Inc16.hdl
/**
* 16-bit incrementer:
* out = in + 1
*/
CHIP Inc16 {
IN in[16];
OUT out[16];
PARTS:
Add16(a=in, b[0]=true, b[1..15]=false, out=out)
}
Chip Inc16 Eval Reset Clock: 0
Input pins
in 1 1 1 1 1 1 1 1 1 1 1 1 1 0 1 1 dec
Output pins
out 1 1 1 1 1 1 1 1 1 1 1 1 1 1 0 0 dec
Test Steps: 7 Outputs: 5
Slow Fast
Test Script Compare File Output File Diff Table
// This file is part of www.nand2tetris.org
// and the book "The Elements of Computing Systems"
// by Nisan and Schocken, MIT Press.
// File name: projects/2/Inc16.tst
load Inc16.hdl,
compare-to Inc16.cmp,
output-list in%B1.16.1 out%B1.16.1;
set in %B0000000000000000, // in = 0
eval,
output;
set in %B1111111111111111, // in = -1
eval,
output;
set in %B0000000000000101, // in = 5
eval,
output;
set in %B1111111111111011, // in = -5
eval,
output;
Simulation successful: The output file is identical to the compare file




```
/**
 * In addition, computes the two output bits:
 * if (out == 0) zr = 1, else zr = 0
 * if (out < 0)  ng = 1, else ng = 0
 */
// Implementation: Manipulates the x and y inputs
// and operates on the resulting values, as follows:
// if (zx == 1) sets x = 0        // 16-bit constant
// if (nx == 1) sets x = !x       // bitwise not
// if (zy == 1) sets y = 0        // 16-bit constant
// if (ny == 1) sets y = !y       // bitwise not
// if (f == 1)  sets out = x + y  // integer 2's complement addition
// if (f == 0)  sets out = x & y  // bitwise and
// if (no == 1) sets out = !out   // bitwise not

CHIP ALU {
IN
x[16], y[16], // 16-bit inputs
zx, // zero the x input?
nx, // negate the x input?
zy, // zero the y input?
ny, // negate the y input?
f, // compute (out = x + y) or (out = x & y)?
no; // negate the out output?
OUT
out[16], // 16-bit output
zr, ng; // if (out == 0) equals 1, else 0
// if (out < 0) equals 1, else 0
PARTS:
Mux16(a=x, b=false, sel=zx, out=xZeroed);
Not16(in=xZeroed, out=notXZeroed);
Mux16(a=xZeroed, b=notXZeroed, sel=nx, out=xProcessed);
Mux16(a=y, b=false, sel=zy, out=yZeroed);
Not16(in=yZeroed, out=notYZeroed);
Mux16(a=yZeroed, b=notYZeroed, sel=ny, out=yProcessed);
Add16(a=xProcessed, b=yProcessed, out=xPlusY);
And16(a=xProcessed, b=yProcessed, out=xAndY);
Mux16(a=xAndY, b=xPlusY, sel=f, out=fResult);
Not16(in=fResult, out=notFResult);
Mux16(a=fResult, b=notFResult, sel=no, out=out, out[0..7]=outLow, out[
Or8Way(in=outLow, out=lowOr);
Or8Way(in=outHigh, out=highOr);
Or(a=lowOr, b=highOr, out=notZr);
Not(in=notZr, out=zr);
}
```

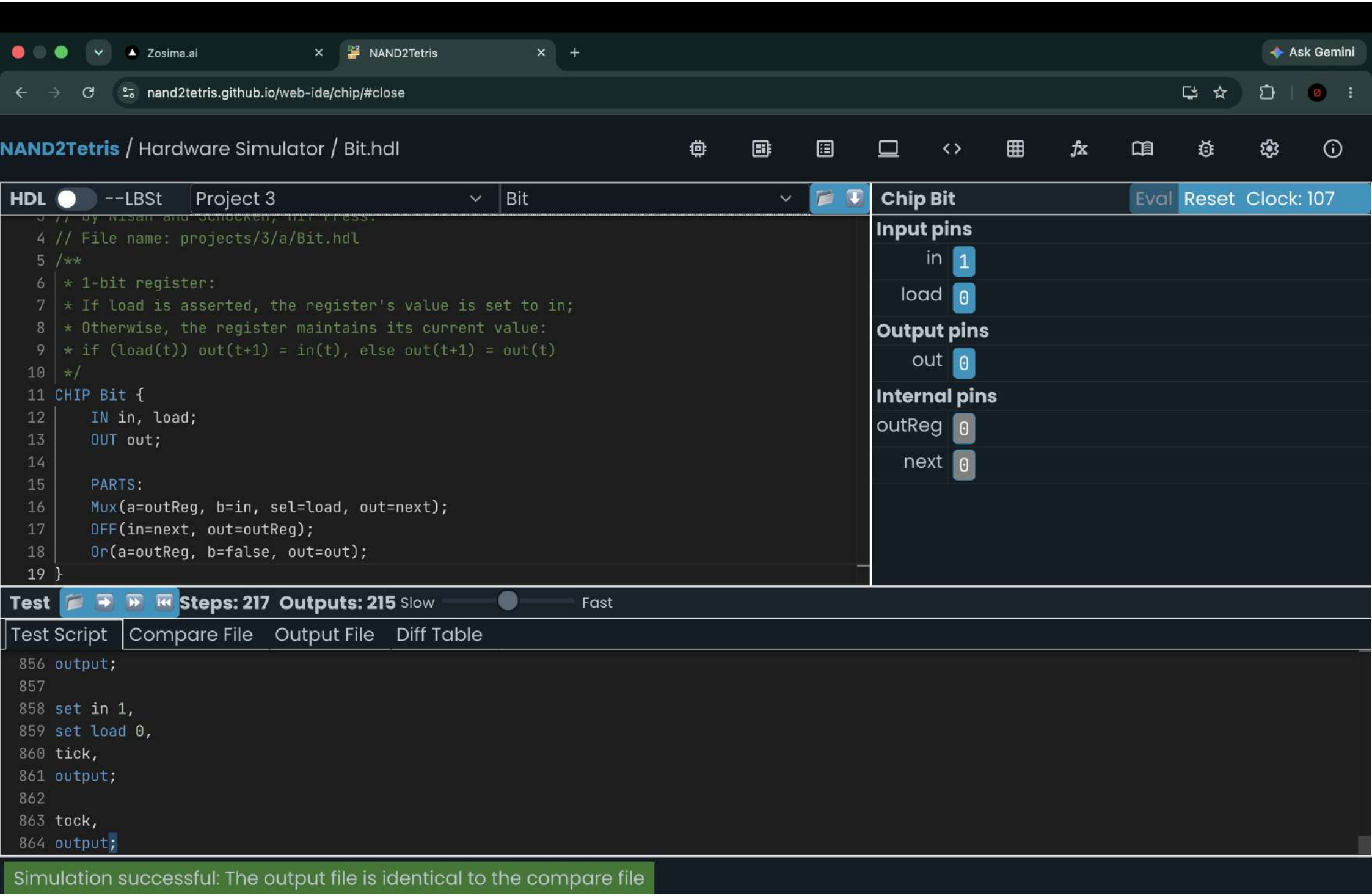
```
set nx 1,
set zy 0,
set ny 0,
set f  1,
set no 1,
eval,
output;

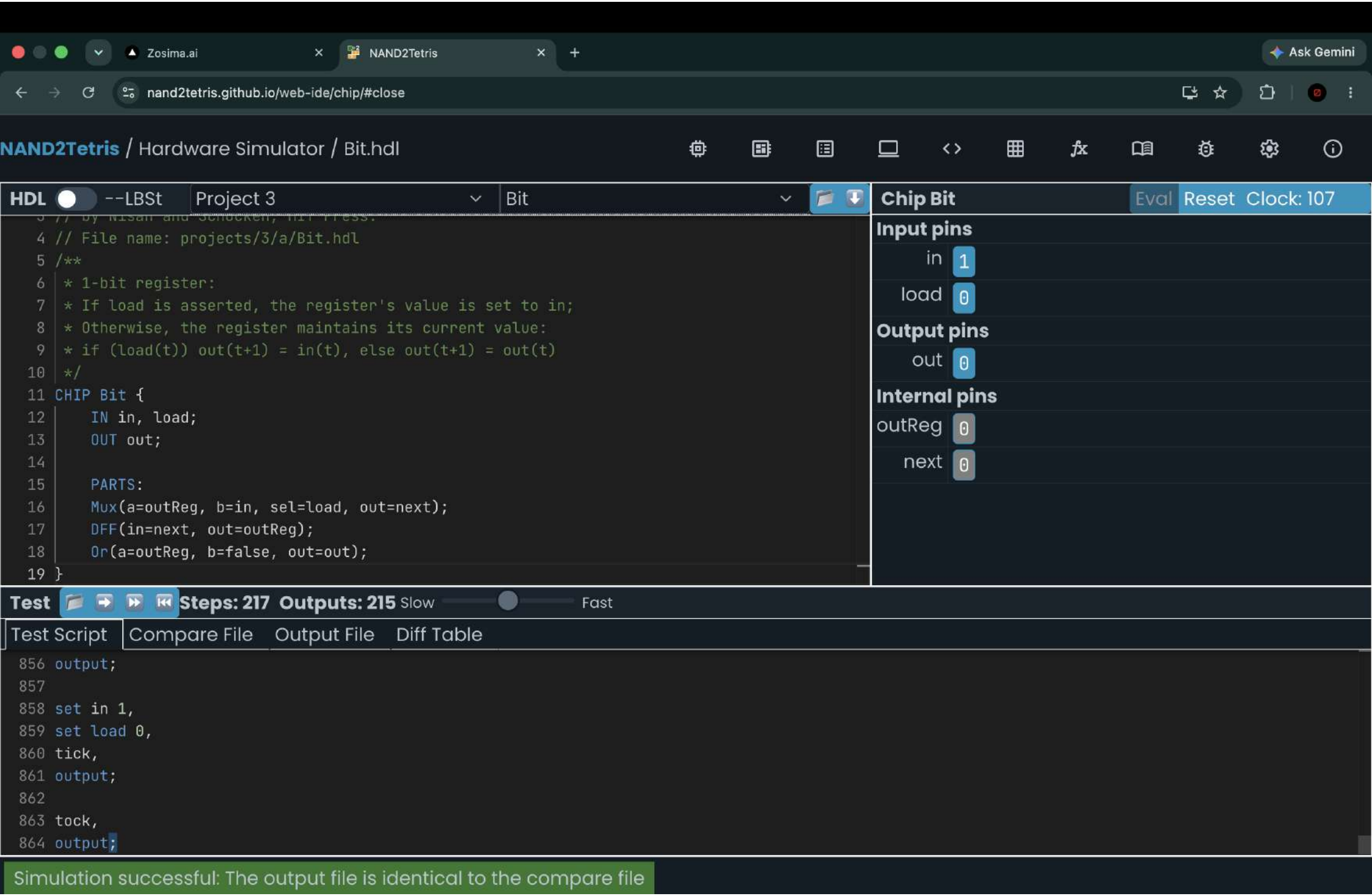
// Compute x + 1
set zx 0,
set nx 1,
set zy 1,
set ny 1,
set f  1,
set no 1,
eval,
output;

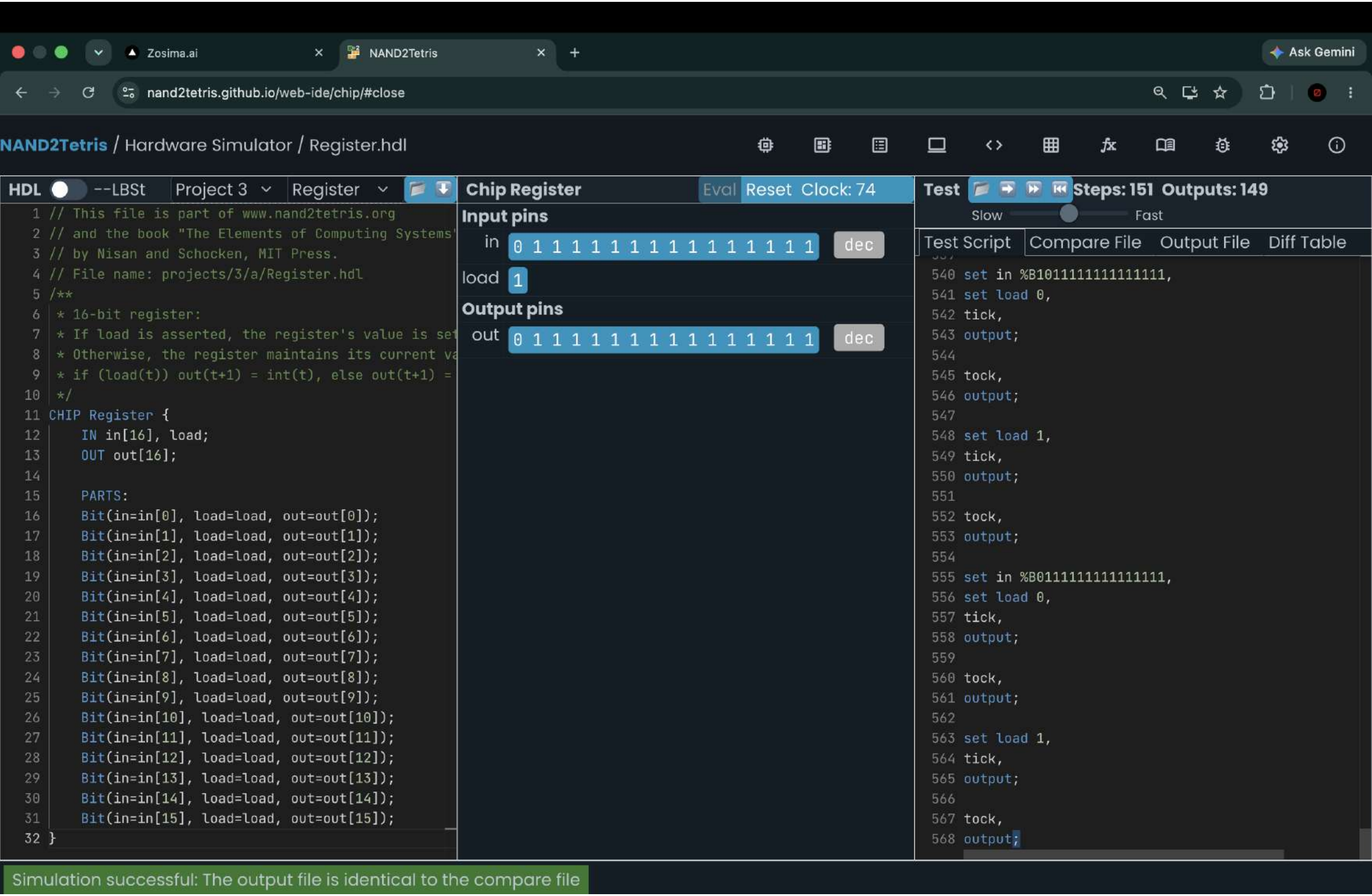
// Compute y + 1
set zx 1,
set nx 1,
set zy 0,
set ny 1,
set f  1,
set no 1,
eval,
output;

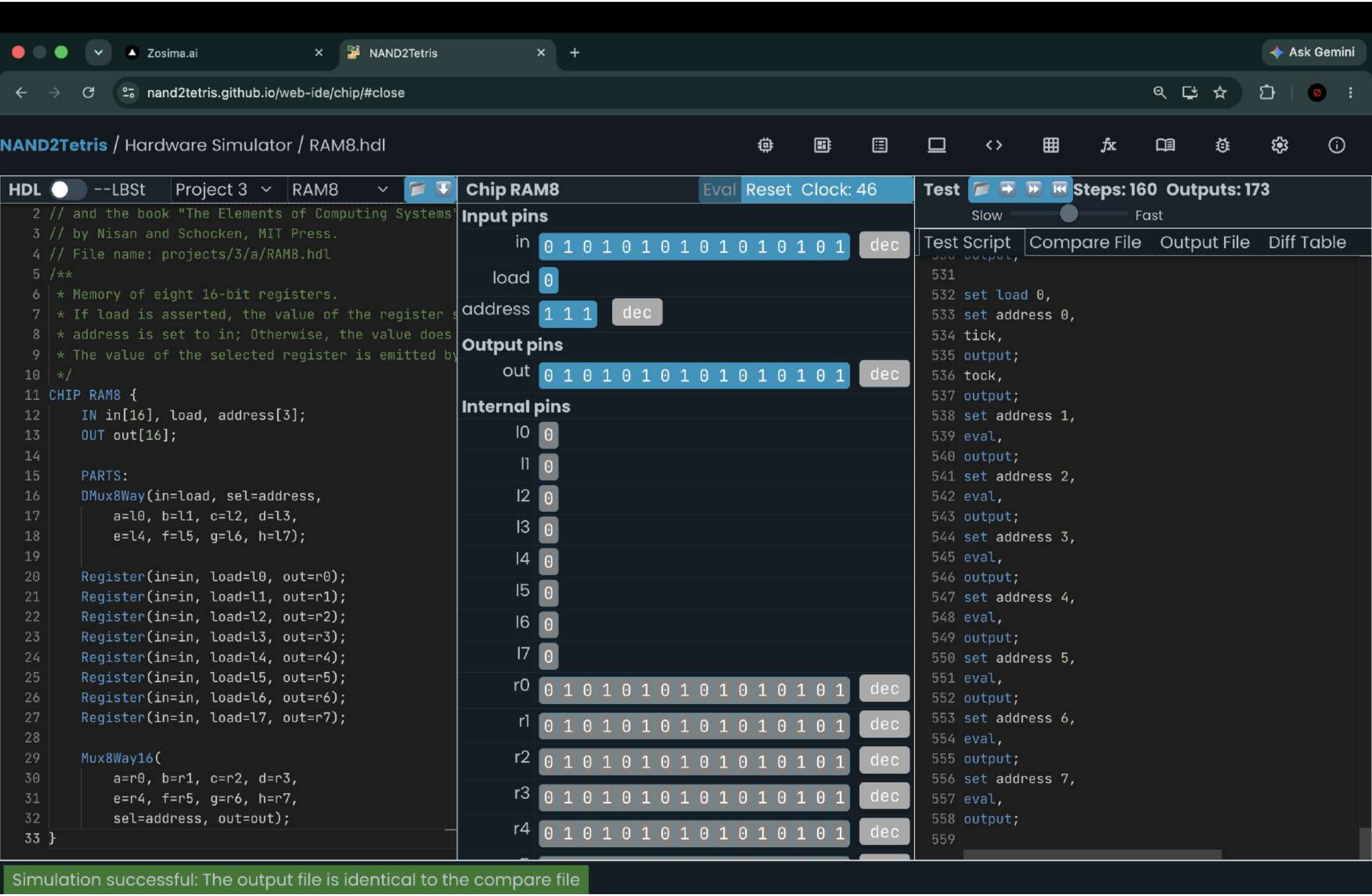
// Compute x - 1
set zx 0,
set nx 0,
set zy 1,
set ny 1,
set f  1,
set no 0,
eval,
output;

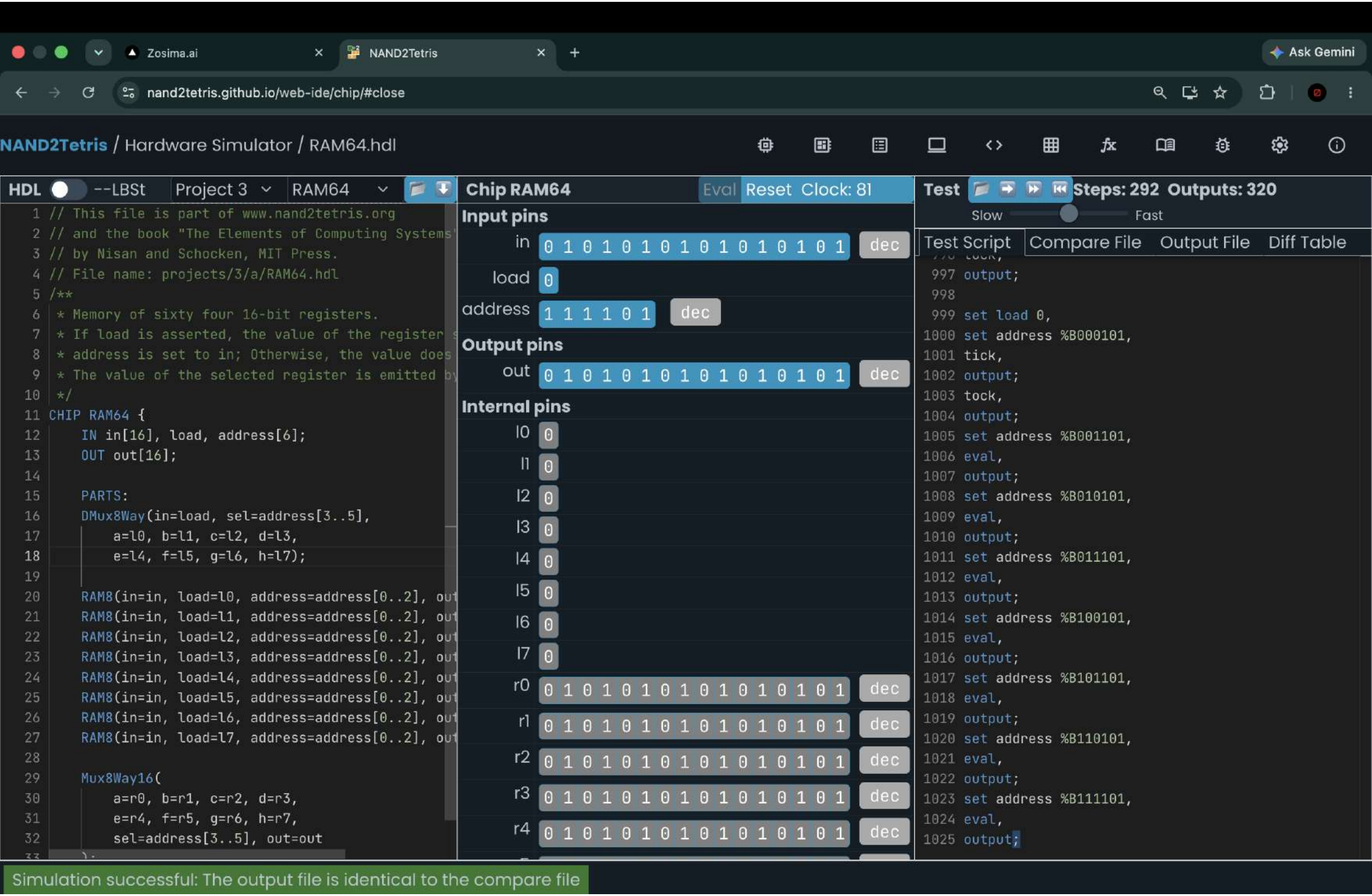
// Compute y - 1
set zx 1,
set nx 1,
set zy 0,
set ny 0,
set f  1,
set no 0,
```

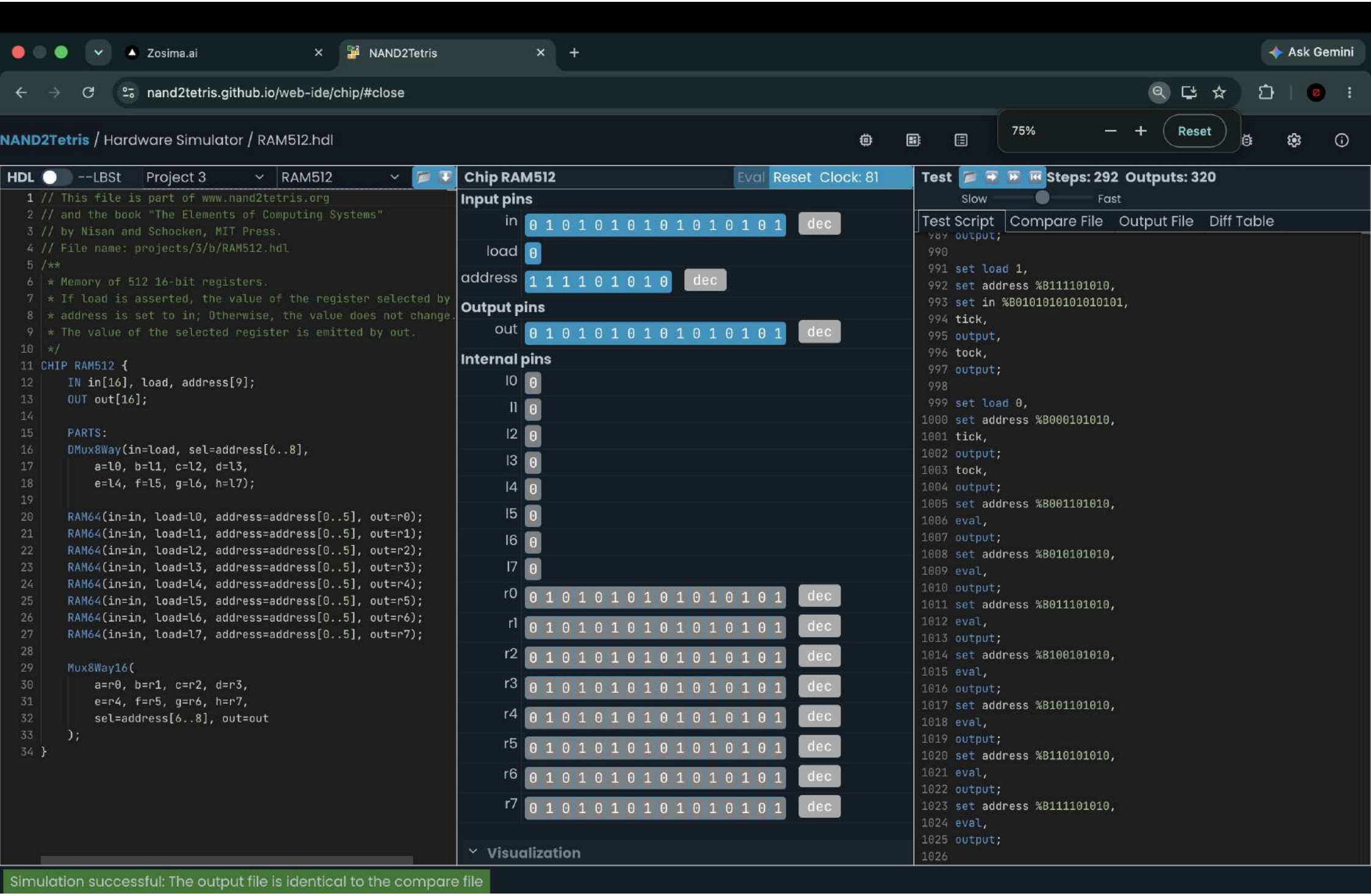

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NAND2Tetris / Hardware Simulator / Bit.hdl
HDL --LBSt
Project 3
Bit
// File name: projects/3/a/Bit.hdl
/**
* 1-bit register:
* If load is asserted, the register's value is set to in;
* Otherwise, the register maintains its current value:
* if (load(t)) out(t+1) = in(t), else out(t+1) = out(t)
*/
CHIP Bit {
IN in, load;
OUT out;
PARTS:
Mux(a=outReg, b=in, sel=load, out=next);
DFF(in=next, out=outReg);
Or(a=outReg, b=false, out=out);
}
Chip Bit
Eval
Reset
Clock: 107
Input pins
in 1
load 0
Output pins
out 0
Internal pins
outReg 0
next 0
Test
Steps: 217 Outputs: 215
Slow
Fast
Test Script
Compare File
Output File
Diff Table
output;
set in 1,
set load 0,
tick,
output;
tock,
output;
Simulation successful: The output file is identical to the compare file

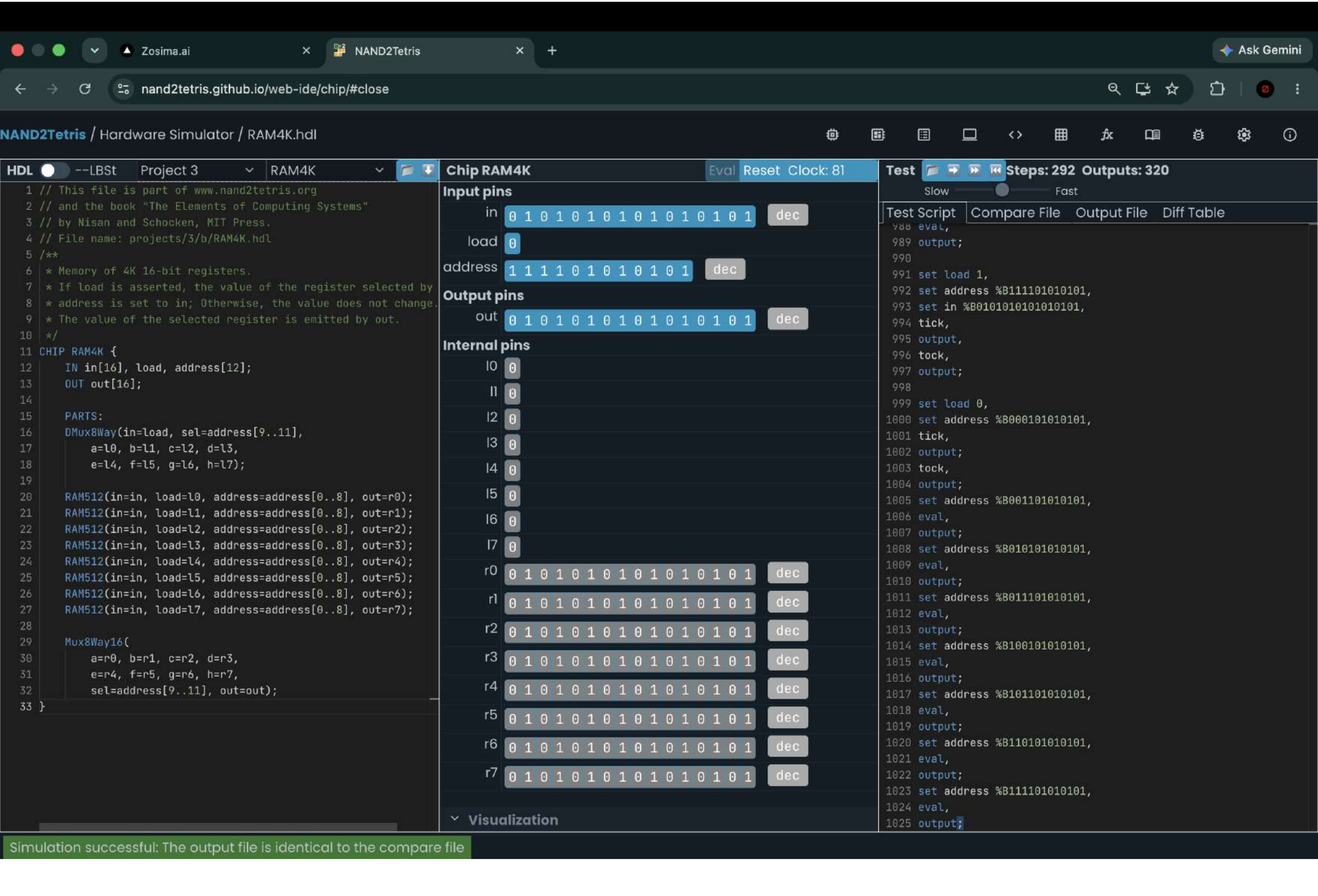

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NAND2Tetris / Hardware Simulator / Bit.hdl
HDL --LBSt
Project 3
Bit
// File name: projects/3/a/Bit.hdl
/**
* 1-bit register:
* If load is asserted, the register's value is set to in;
* Otherwise, the register maintains its current value:
* if (load(t)) out(t+1) = in(t), else out(t+1) = out(t)
*/
CHIP Bit {
IN in, load;
OUT out;
PARTS:
Mux(a=outReg, b=in, sel=load, out=next);
DFF(in=next, out=outReg);
Or(a=outReg, b=false, out=out);
}
Chip Bit
Eval
Reset
Clock: 107
Input pins
in 1
load 0
Output pins
out 0
Internal pins
outReg 0
next 0
Test
Steps: 217 Outputs: 215
Slow
Fast
Test Script
Compare File
Output File
Diff Table
output;
set in 1,
set load 0,
tick,
output;
tock,
output;
Simulation successful: The output file is identical to the compare file

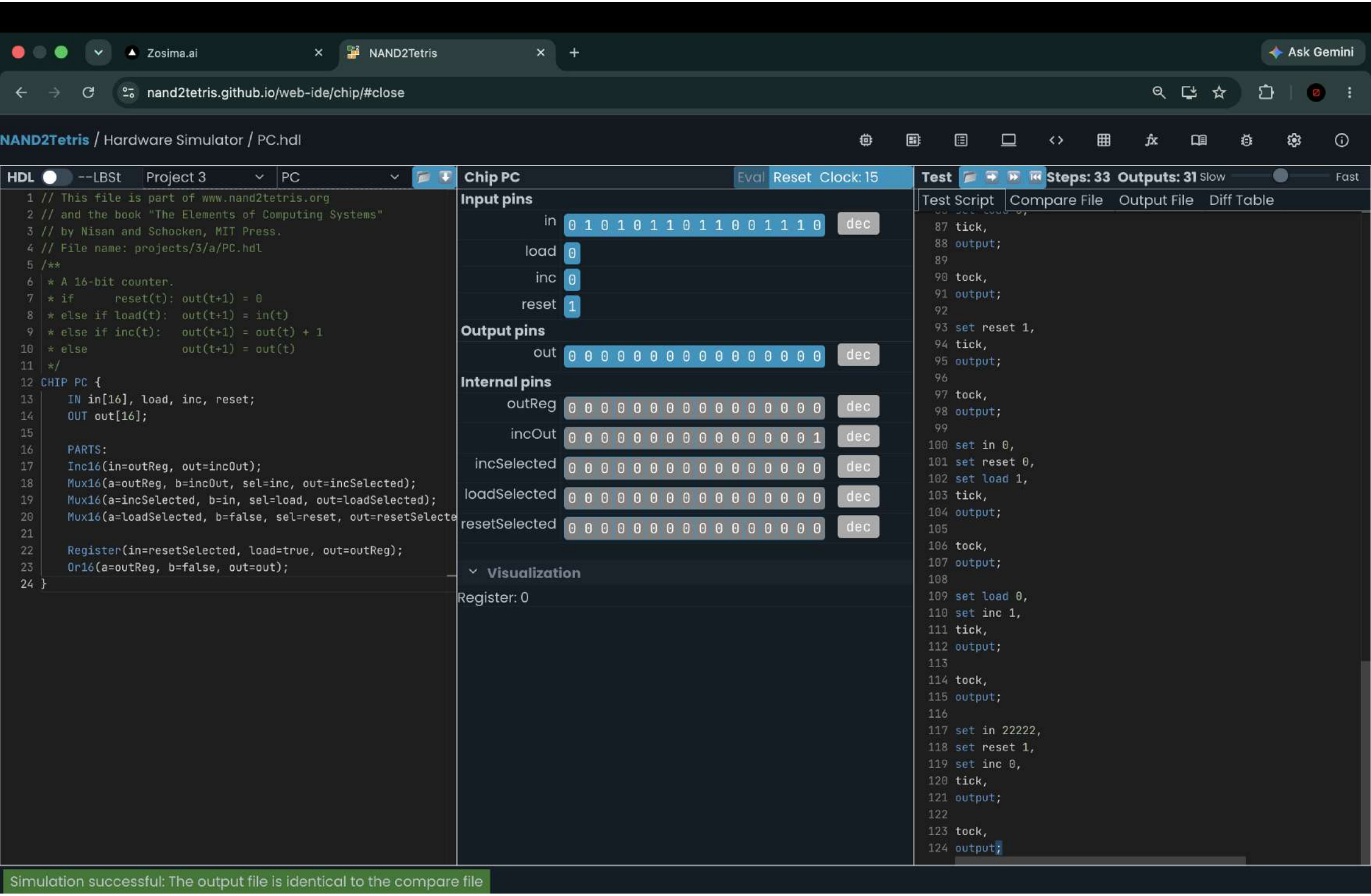

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NAND2Tetris / Hardware Simulator / Register.hdl
HDL --LBSt Project 3 Register
// This file is part of www.nand2tetris.org
// and the book "The Elements of Computing Systems"
// by Nisan and Schocken, MIT Press.
// File name: projects/3/a/Register.hdl
/**
* 16-bit register:
* If load is asserted, the register's value is set
* Otherwise, the register maintains its current va
* if (load(t)) out(t+1) = int(t), else out(t+1) =
*/
CHIP Register {
IN in[16], load;
OUT out[16];
PARTS:
Bit(in=in[0], load=load, out=out[0]);
Bit(in=in[1], load=load, out=out[1]);
Bit(in=in[2], load=load, out=out[2]);
Bit(in=in[3], load=load, out=out[3]);
Bit(in=in[4], load=load, out=out[4]);
Bit(in=in[5], load=load, out=out[5]);
Bit(in=in[6], load=load, out=out[6]);
Bit(in=in[7], load=load, out=out[7]);
Bit(in=in[8], load=load, out=out[8]);
Bit(in=in[9], load=load, out=out[9]);
Bit(in=in[10], load=load, out=out[10]);
Bit(in=in[11], load=load, out=out[11]);
Bit(in=in[12], load=load, out=out[12]);
Bit(in=in[13], load=load, out=out[13]);
Bit(in=in[14], load=load, out=out[14]);
Bit(in=in[15], load=load, out=out[15]);
}
Chip Register
Eval Reset Clock: 74
Input pins
in 0 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 dec
load 1
Output pins
out 0 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 dec
Test
Steps: 151 Outputs: 149
Slow Fast
Test Script Compare File Output File Diff Table
set in %B1011111111111111,
set load 0,
tick,
output;
tock,
output;
set load 1,
tick,
output;
tock,
output;
set in %B0111111111111111,
set load 0,
tick,
output;
tock,
output;
set load 1,
tick,
output;
tock,
output;
Simulation successful: The output file is identical to the compare file

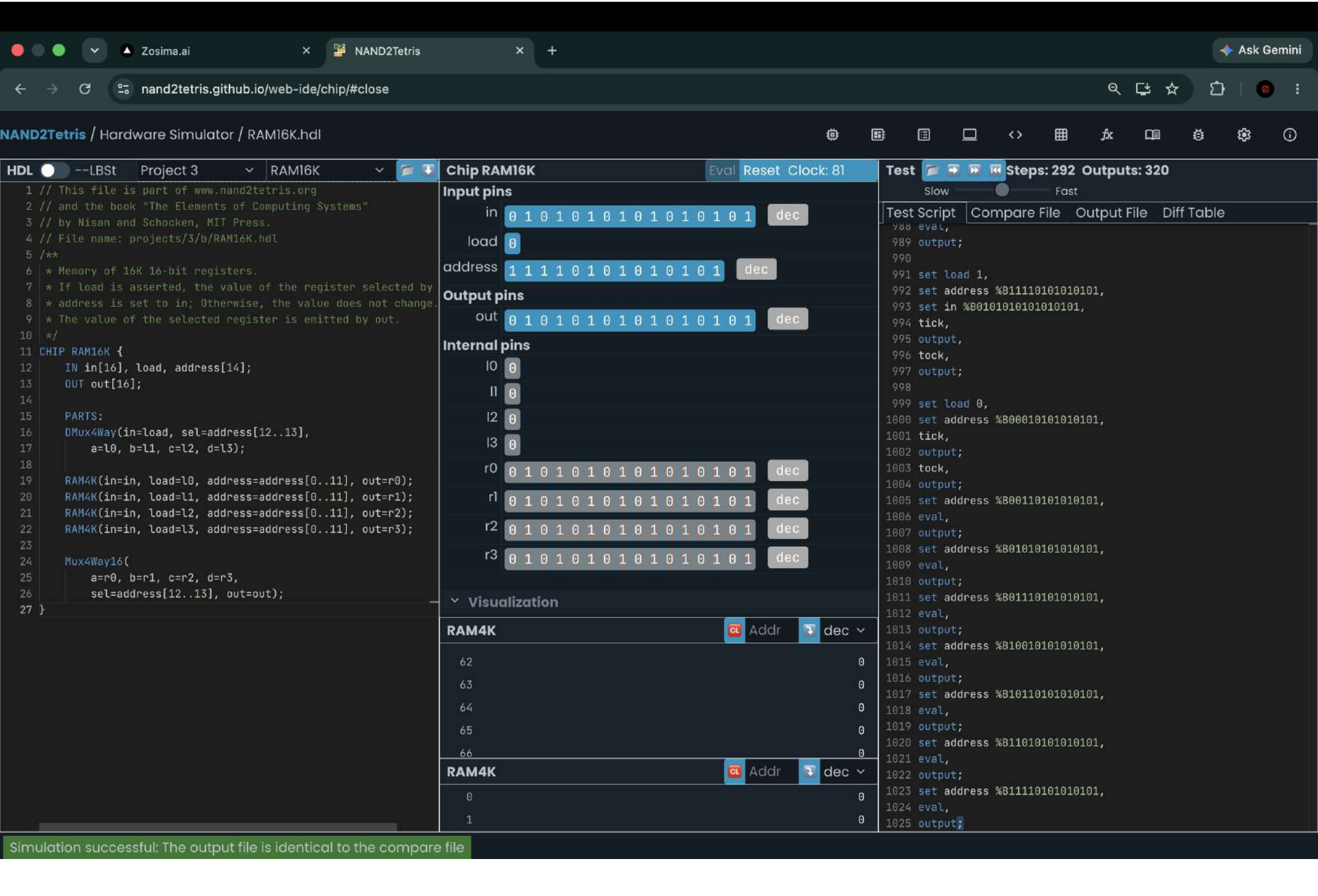

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NAND2Tetris / Hardware Simulator / RAM8.hdl
HDL --LBSt Project 3 RAM8
// and the book "The Elements of Computing Systems"
// by Nisan and Schocken, MIT Press.
// File name: projects/3/a/RAM8.hdl
/**
* Memory of eight 16-bit registers.
* If load is asserted, the value of the register s
* address is set to in; Otherwise, the value does
* The value of the selected register is emitted by
*/
CHIP RAM8 {
IN in[16], load, address[3];
OUT out[16];
PARTS:
DMux8Way(in=load, sel=address,
a=l0, b=l1, c=l2, d=l3,
e=l4, f=l5, g=l6, h=l7);
Register(in=in, load=l0, out=r0);
Register(in=in, load=l1, out=r1);
Register(in=in, load=l2, out=r2);
Register(in=in, load=l3, out=r3);
Register(in=in, load=l4, out=r4);
Register(in=in, load=l5, out=r5);
Register(in=in, load=l6, out=r6);
Register(in=in, load=l7, out=r7);
Mux8Way16(
a=r0, b=r1, c=r2, d=r3,
e=r4, f=r5, g=r6, h=r7,
sel=address, out=out);
}
Chip RAM8 Eval Reset Clock: 46
Input pins
in 0 1 0 1 0 1 0 1 0 1 0 1 0 1 0 1 dec
load 0
address 1 1 1 dec
Output pins
out 0 1 0 1 0 1 0 1 0 1 0 1 0 1 0 1 dec
Internal pins
l0 0
l1 0
l2 0
l3 0
l4 0
l5 0
l6 0
l7 0
r0 0 1 0 1 0 1 0 1 0 1 0 1 0 1 0 1 dec
r1 0 1 0 1 0 1 0 1 0 1 0 1 0 1 0 1 dec
r2 0 1 0 1 0 1 0 1 0 1 0 1 0 1 0 1 dec
r3 0 1 0 1 0 1 0 1 0 1 0 1 0 1 0 1 dec
r4 0 1 0 1 0 1 0 1 0 1 0 1 0 1 0 1 dec
Test Steps: 160 Outputs: 173
Slow Fast
Test Script Compare File Output File Diff Table
set load 0,
set address 0,
tick,
output;
tock,
output;
set address 1,
eval,
output;
set address 2,
eval,
output;
set address 3,
eval,
output;
set address 4,
eval,
output;
set address 5,
eval,
output;
set address 6,
eval,
output;
set address 7,
eval,
output;
Simulation successful: The output file is identical to the compare file


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NAND2Tetris / Hardware Simulator / RAM64.hdl
HDL --LBSt Project 3 RAM64
// This file is part of www.nand2tetris.org
// and the book "The Elements of Computing Systems"
// by Nisan and Schocken, MIT Press.
// File name: projects/3/a/RAM64.hdl
/**
* Memory of sixty four 16-bit registers.
* If load is asserted, the value of the register
* address is set to in; Otherwise, the value does
* The value of the selected register is emitted by
*/
CHIP RAM64 {
IN in[16], load, address[6];
OUT out[16];
PARTS:
DMux8Way(in=load, sel=address[3..5],
a=l0, b=l1, c=l2, d=l3,
e=l4, f=l5, g=l6, h=l7);
RAM8(in=in, load=l0, address=address[0..2], out
RAM8(in=in, load=l1, address=address[0..2], out
RAM8(in=in, load=l2, address=address[0..2], out
RAM8(in=in, load=l3, address=address[0..2], out
RAM8(in=in, load=l4, address=address[0..2], out
RAM8(in=in, load=l5, address=address[0..2], out
RAM8(in=in, load=l6, address=address[0..2], out
RAM8(in=in, load=l7, address=address[0..2], out
Mux8Way16(
a=r0, b=r1, c=r2, d=r3,
e=r4, f=r5, g=r6, h=r7,
sel=address[3..5], out=out
);
Chip RAM64 Eval Reset Clock: 81
Input pins
in 0 1 0 1 0 1 0 1 0 1 0 1 0 1 0 1 dec
load 0
address 1 1 1 1 0 1 dec
Output pins
out 0 1 0 1 0 1 0 1 0 1 0 1 0 1 0 1 dec
Internal pins
l0 0
l1 0
l2 0
l3 0
l4 0
l5 0
l6 0
l7 0
r0 0 1 0 1 0 1 0 1 0 1 0 1 0 1 0 1 dec
r1 0 1 0 1 0 1 0 1 0 1 0 1 0 1 0 1 dec
r2 0 1 0 1 0 1 0 1 0 1 0 1 0 1 0 1 dec
r3 0 1 0 1 0 1 0 1 0 1 0 1 0 1 0 1 dec
r4 0 1 0 1 0 1 0 1 0 1 0 1 0 1 0 1 dec
Test Steps: 292 Outputs: 320
Slow Fast
Test Script Compare File Output File Diff Table
997 output;
998
999 set load 0,
1000 set address %B000101,
1001 tick,
1002 output;
1003 tock,
1004 output;
1005 set address %B001101,
1006 eval,
1007 output;
1008 set address %B010101,
1009 eval,
1010 output;
1011 set address %B011101,
1012 eval,
1013 output;
1014 set address %B100101,
1015 eval,
1016 output;
1017 set address %B101101,
1018 eval,
1019 output;
1020 set address %B110101,
1021 eval,
1022 output;
1023 set address %B111101,
1024 eval,
1025 output;
Simulation successful: The output file is identical to the compare file


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NAND2Tetris / Hardware Simulator / RAM512.hdl
75%
Reset
HDL
--LBSt
Project 3
RAM512
// This file is part of www.nand2tetris.org
// and the book "The Elements of Computing Systems"
// by Nisan and Schocken, MIT Press.
// File name: projects/3/b/RAM512.hdl
/**
* Memory of 512 16-bit registers.
* If load is asserted, the value of the register selected by
* address is set to in; Otherwise, the value does not change.
* The value of the selected register is emitted by out.
*/
CHIP RAM512 {
IN in[16], load, address[9];
OUT out[16];
PARTS:
DMux8Way(in=load, sel=address[6..8],
a=l0, b=l1, c=l2, d=l3,
e=l4, f=l5, g=l6, h=l7);
RAM64(in=in, load=l0, address=address[0..5], out=r0);
RAM64(in=in, load=l1, address=address[0..5], out=r1);
RAM64(in=in, load=l2, address=address[0..5], out=r2);
RAM64(in=in, load=l3, address=address[0..5], out=r3);
RAM64(in=in, load=l4, address=address[0..5], out=r4);
RAM64(in=in, load=l5, address=address[0..5], out=r5);
RAM64(in=in, load=l6, address=address[0..5], out=r6);
RAM64(in=in, load=l7, address=address[0..5], out=r7);
Mux8Way16(
a=r0, b=r1, c=r2, d=r3,
e=r4, f=r5, g=r6, h=r7,
sel=address[6..8], out=out
);
}
Chip RAM512
Eval
Reset
Clock: 81
Input pins
in 0 1 0 1 0 1 0 1 0 1 0 1 0 1 0 1 dec
load 0
address 1 1 1 1 0 1 0 1 0 dec
Output pins
out 0 1 0 1 0 1 0 1 0 1 0 1 0 1 0 1 dec
Internal pins
l0 0
l1 0
l2 0
l3 0
l4 0
l5 0
l6 0
l7 0
r0 0 1 0 1 0 1 0 1 0 1 0 1 0 1 0 1 dec
r1 0 1 0 1 0 1 0 1 0 1 0 1 0 1 0 1 dec
r2 0 1 0 1 0 1 0 1 0 1 0 1 0 1 0 1 dec
r3 0 1 0 1 0 1 0 1 0 1 0 1 0 1 0 1 dec
r4 0 1 0 1 0 1 0 1 0 1 0 1 0 1 0 1 dec
r5 0 1 0 1 0 1 0 1 0 1 0 1 0 1 0 1 dec
r6 0 1 0 1 0 1 0 1 0 1 0 1 0 1 0 1 dec
r7 0 1 0 1 0 1 0 1 0 1 0 1 0 1 0 1 dec
Visualization
Test
Steps: 292 Outputs: 320
Slow
Fast
Test Script
Compare File
Output File
Diff Table
output;
set load 1,
set address %B111101010,
set in %B0101010101010101,
tick,
output,
tock,
output;
set load 0,
set address %B000101010,
tick,
output;
tock,
output;
set address %B001101010,
eval,
output;
set address %B010101010,
eval,
output;
set address %B011101010,
eval,
output;
set address %B100101010,
eval,
output;
set address %B101101010,
eval,
output;
set address %B110101010,
eval,
output;
set address %B111101010,
eval,
output;
Simulation successful: The output file is identical to the compare file


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NAND2Tetris / Hardware Simulator / RAM4K.hdl
HDL --LBSt Project 3 RAM4K
// This file is part of www.nand2tetris.org
// and the book "The Elements of Computing Systems"
// by Nisan and Schocken, MIT Press.
// File name: projects/3/b/RAM4K.hdl
/**
* Memory of 4K 16-bit registers.
* If load is asserted, the value of the register selected by
* address is set to in; Otherwise, the value does not change.
* The value of the selected register is emitted by out.
*/
CHIP RAM4K {
IN in[16], load, address[12];
OUT out[16];
PARTS:
DMux8Way(in=load, sel=address[9..11],
a=l0, b=l1, c=l2, d=l3,
e=l4, f=l5, g=l6, h=l7);
RAM512(in=in, load=l0, address=address[0..8], out=r0);
RAM512(in=in, load=l1, address=address[0..8], out=r1);
RAM512(in=in, load=l2, address=address[0..8], out=r2);
RAM512(in=in, load=l3, address=address[0..8], out=r3);
RAM512(in=in, load=l4, address=address[0..8], out=r4);
RAM512(in=in, load=l5, address=address[0..8], out=r5);
RAM512(in=in, load=l6, address=address[0..8], out=r6);
RAM512(in=in, load=l7, address=address[0..8], out=r7);
Mux8Way16(
a=r0, b=r1, c=r2, d=r3,
e=r4, f=r5, g=r6, h=r7,
sel=address[9..11], out=out);
}
Chip RAM4K
Eval
Reset
Clock: 81
Input pins
in 0 1 0 1 0 1 0 1 0 1 0 1 0 1 0 1 dec
load 0
address 1 1 1 1 0 1 0 1 0 1 0 1 dec
Output pins
out 0 1 0 1 0 1 0 1 0 1 0 1 0 1 0 1 dec
Internal pins
l0 0
l1 0
l2 0
l3 0
l4 0
l5 0
l6 0
l7 0
r0 0 1 0 1 0 1 0 1 0 1 0 1 0 1 0 1 dec
r1 0 1 0 1 0 1 0 1 0 1 0 1 0 1 0 1 dec
r2 0 1 0 1 0 1 0 1 0 1 0 1 0 1 0 1 dec
r3 0 1 0 1 0 1 0 1 0 1 0 1 0 1 0 1 dec
r4 0 1 0 1 0 1 0 1 0 1 0 1 0 1 0 1 dec
r5 0 1 0 1 0 1 0 1 0 1 0 1 0 1 0 1 dec
r6 0 1 0 1 0 1 0 1 0 1 0 1 0 1 0 1 dec
r7 0 1 0 1 0 1 0 1 0 1 0 1 0 1 0 1 dec
Visualization
Test
Steps: 292 Outputs: 320
Slow
Fast
Test Script
Compare File
Output File
Diff Table
988 eval,
989 output;
990
991 set load 1,
992 set address %B111101010101,
993 set in %B0101010101010101,
994 tick,
995 output,
996 tock,
997 output;
998
999 set load 0,
1000 set address %B000101010101,
1001 tick,
1002 output;
1003 tock,
1004 output;
1005 set address %B001101010101,
1006 eval,
1007 output;
1008 set address %B010101010101,
1009 eval,
1010 output;
1011 set address %B011101010101,
1012 eval,
1013 output;
1014 set address %B100101010101,
1015 eval,
1016 output;
1017 set address %B101101010101,
1018 eval,
1019 output;
1020 set address %B110101010101,
1021 eval,
1022 output;
1023 set address %B111101010101,
1024 eval,
1025 output;
Simulation successful: The output file is identical to the compare file


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NAND2Tetris / Hardware Simulator / PC.hdl
HDL --LBSt Project 3 PC
// This file is part of www.nand2tetris.org
// and the book "The Elements of Computing Systems"
// by Nisan and Schocken, MIT Press.
// File name: projects/3/a/PC.hdl
/**
* A 16-bit counter.
* if reset(t): out(t+1) = 0
* else if load(t): out(t+1) = in(t)
* else if inc(t): out(t+1) = out(t) + 1
* else out(t+1) = out(t)
*/
CHIP PC {
IN in[16], load, inc, reset;
OUT out[16];
PARTS:
Inc16(in=outReg, out=incOut);
Mux16(a=outReg, b=incOut, sel=inc, out=incSelected);
Mux16(a=incSelected, b=in, sel=load, out=loadSelected);
Mux16(a=loadSelected, b=false, sel=reset, out=resetSelecte
Register(in=resetSelected, load=true, out=outReg);
Or16(a=outReg, b=false, out=out);
}
Chip PC Eval Reset Clock: 15
Input pins
in 0 1 0 1 0 1 1 0 1 1 0 0 1 1 1 0 dec
load 0
inc 0
reset 1
Output pins
out 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 dec
Internal pins
outReg 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 dec
incOut 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 1 dec
incSelected 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 dec
loadSelected 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 dec
resetSelected 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 dec
Visualization
Register: 0
Test Steps: 33 Outputs: 31 Slow Fast
Test Script Compare File Output File Diff Table
87 tick,
88 output;
89
90 tock,
91 output;
92
93 set reset 1,
94 tick,
95 output;
96
97 tock,
98 output;
99
100 set in 0,
101 set reset 0,
102 set load 1,
103 tick,
104 output;
105
106 tock,
107 output;
108
109 set load 0,
110 set inc 1,
111 tick,
112 output;
113
114 tock,
115 output;
116
117 set in 22222,
118 set reset 1,
119 set inc 0,
120 tick,
121 output;
122
123 tock,
124 output;
Simulation successful: The output file is identical to the compare file


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NAND2Tetris / Hardware Simulator / RAM16K.hdl
HDL --LBSt Project 3 RAM16K
1 // This file is part of www.nand2tetris.org
2 // and the book "The Elements of Computing Systems"
3 // by Nisan and Schocken, MIT Press.
4 // File name: projects/3/b/RAM16K.hdl
5 /**
6 * Memory of 16K 16-bit registers.
7 * If load is asserted, the value of the register selected by
8 * address is set to in; Otherwise, the value does not change.
9 * The value of the selected register is emitted by out.
10 */
11 CHIP RAM16K {
12 IN in[16], load, address[14];
13 OUT out[16];
14
15 PARTS:
16 DMux4Way(in=load, sel=address[12..13],
17 a=l0, b=l1, c=l2, d=l3);
18
19 RAM4K(in=in, load=l0, address=address[0..11], out=r0);
20 RAM4K(in=in, load=l1, address=address[0..11], out=r1);
21 RAM4K(in=in, load=l2, address=address[0..11], out=r2);
22 RAM4K(in=in, load=l3, address=address[0..11], out=r3);
23
24 Mux4Way16(
25 a=r0, b=r1, c=r2, d=r3,
26 sel=address[12..13], out=out);
27 }
Chip RAM16K Eval Reset Clock: 81
Input pins
in 0 1 0 1 0 1 0 1 0 1 0 1 0 1 0 1 dec
load 0
address 1 1 1 1 0 1 0 1 0 1 0 1 0 1 dec
Output pins
out 0 1 0 1 0 1 0 1 0 1 0 1 0 1 0 1 dec
Internal pins
l0 0
l1 0
l2 0
l3 0
r0 0 1 0 1 0 1 0 1 0 1 0 1 0 1 0 1 dec
r1 0 1 0 1 0 1 0 1 0 1 0 1 0 1 0 1 dec
r2 0 1 0 1 0 1 0 1 0 1 0 1 0 1 0 1 dec
r3 0 1 0 1 0 1 0 1 0 1 0 1 0 1 0 1 dec
Visualization
RAM4K CL Addr dec
62 0
63 0
64 0
65 0
66 0
RAM4K CL Addr dec
0 0
1 0
Test Steps: 292 Outputs: 320
Slow Fast
Test Script Compare File Output File Diff Table
988 eval,
989 output;
990
991 set load 1,
992 set address %B11110101010101,
993 set in %B0101010101010101,
994 tick,
995 output,
996 tock,
997 output;
998
999 set load 0,
1000 set address %B00010101010101,
1001 tick,
1002 output;
1003 tock,
1004 output;
1005 set address %B00110101010101,
1006 eval,
1007 output;
1008 set address %B01010101010101,
1009 eval,
1010 output;
1011 set address %B01110101010101,
1012 eval,
1013 output;
1014 set address %B10010101010101,
1015 eval,
1016 output;
1017 set address %B10110101010101,
1018 eval,
1019 output;
1020 set address %B11010101010101,
1021 eval,
1022 output;
1023 set address %B11110101010101,
1024 eval,
1025 output;
Simulation successful: The output file is identical to the compare file